# 2. A (Quick) Introduction to Cryptography

Computer Security Courses @ POLIMI

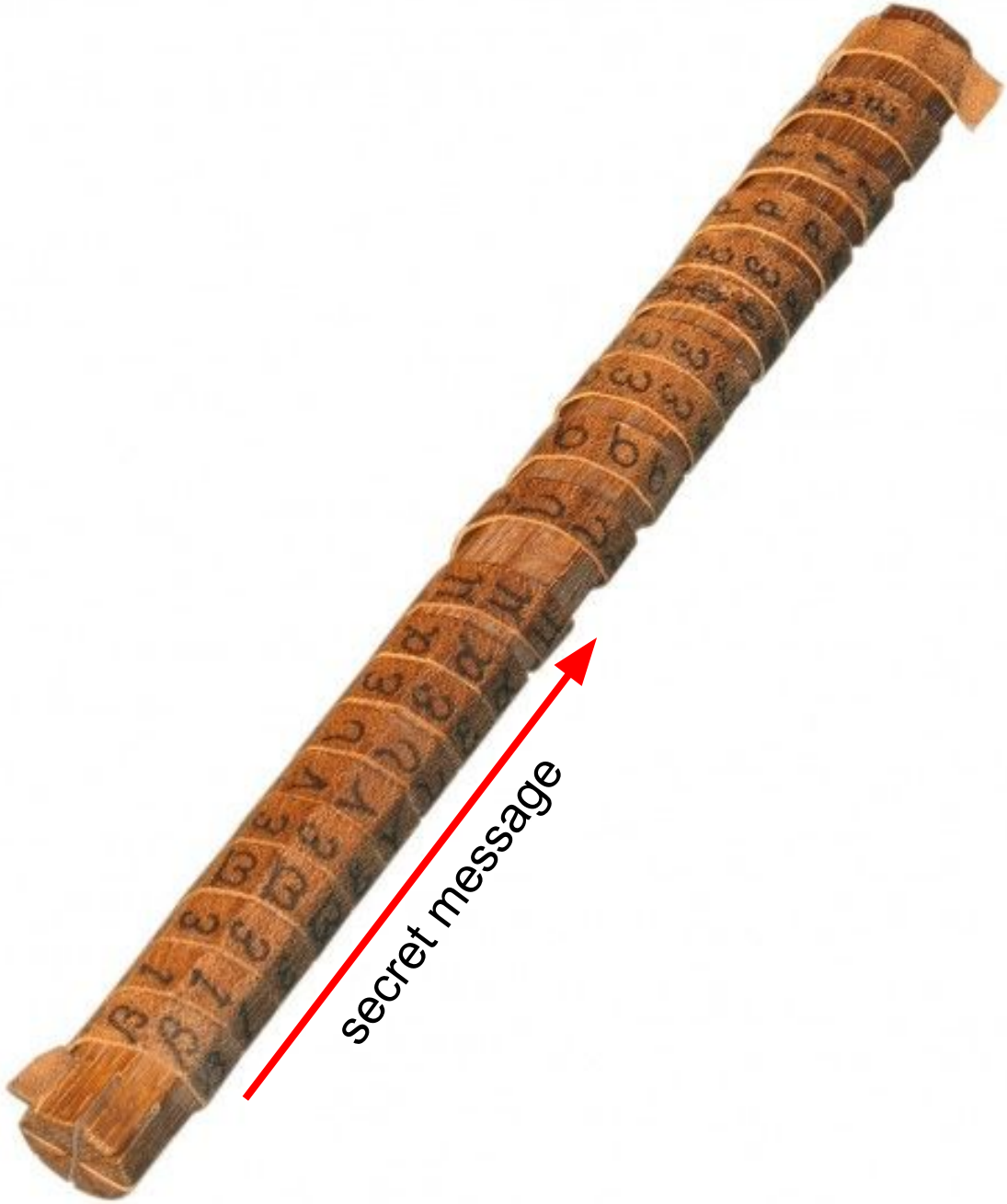
*Prof. Carminati & Prof. Zanero*

# Word of Warning

- This is a short, simplified introduction to cryptography
- We will only introduce what is needed for systems security discussions
- Mostly, we will treat mathematical concepts as black boxes
- A much more in-depth discussion of cryptography takes place in course:
  - 095947 - CRYPTOGRAPHY AND ARCHITECTURES FOR COMPUTER SECURITY (Prof. G. Pelosi)

# A Brief History of Cryptography (1)

- From Greek: *kryptos*, hidden, and *graphein*, to write (i.e., *"art of secret writing"*)
- Ancient history: writing itself was already a "secret technique".
- Cryptography born in Greek society, when writing became more common, and hidden writing became a need.
  - E.g. "scytale", the wand of command of the army of Athens. According to Plutarco, in use since the IX century b.C.

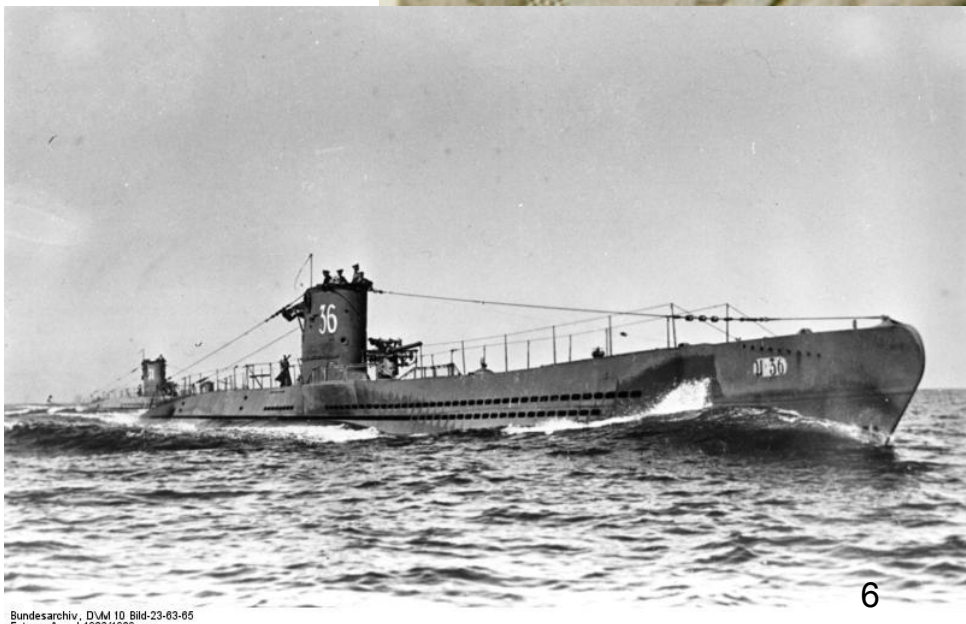

secret message

# A Brief History of Cryptography (2)

- Medieval and renaissance studies
  - **Gabriele de Lavinde**, who wrote a manual in 1379, copy available at the Vatican archives.
  - The mirror writing of **Leonardo da Vinci**.
- Mostly a *military interest*
  - Italian Army General **Luigi Sacco** wrote a famous *"Nozioni di crittografia"* book in 1925, one of the last "non-formalized" exercises in cryptography.
- Also the formalisms of cryptography were born out of wartime needs.

# When Math Won a War

- During WW II, **Alan Turing** worked at Bletchley Park to **break** Axis ciphers, in particular the **Enigma cipher**.
- Birth of the first universal computers was stimulated by this effort.




Bundesarchiv, D\VM 10 Bild-23-63-65
Foto: o.Ang. | 1936/1939 ca.

# When Math Won a War

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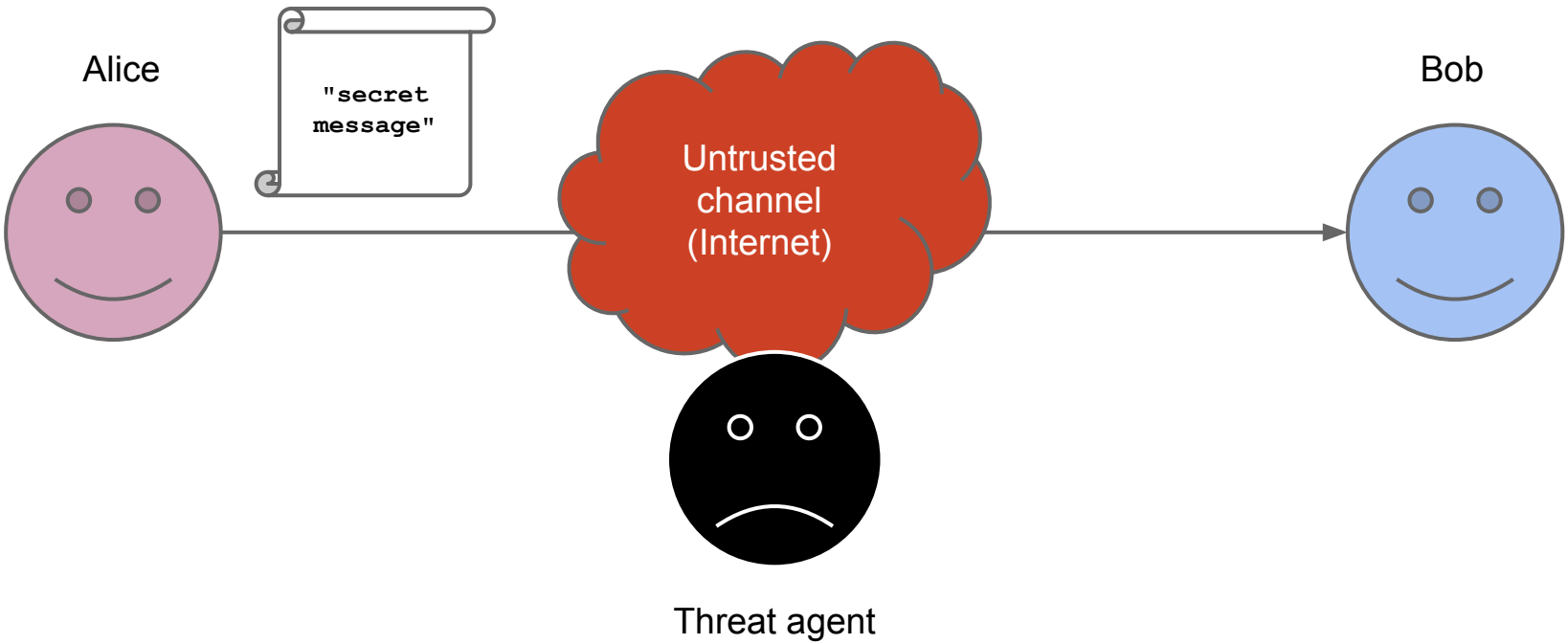
THE BOMBE (replica)

# Key Concepts in Cryptography

- First formalized by Claude Shannon in his 1949 paper *"Communication theory of secrecy systems"*.
- **Cryptosystem**: a system that takes in input a message (known as **plaintext**) and transforms it into a **ciphertext** with a reversible function that usually takes a **key** as a further input.
- The use of *"text"* is historical, and today we mean *"string of bits"*.

# The Problem to Solve: Confidentiality and Integrity



# Kerckhoffs' Principle

- *The security of a cryptosystem relies only on the secrecy of the key, and never on the secrecy of the algorithm.*
  - Auguste Kerckhoffs, *“La criptographie militaire”*, 1883
  - Sometimes called “Kerchoffs/Shannon”
- This means that:
  - In a secure cryptosystem we cannot retrieve the plaintext from the ciphertext without the key.
  - Also, we cannot retrieve the key from analyzing ciphertext-plaintext pairs.
  - Algorithms must always be assumed known to the attacker, no secret sauce!

# Perfection is not of This World

Shannon wondered: *"Is there a perfect cipher?"*

A cipher such that no matter time or strength spent in analysis, does not leak any secret?

- $P(M = m)$ probability of observing message $m$
- $P(M = m \mid C = c)$ probability that message was *m* | given that ciphertext *c* was observed

The cipher is a **perfect cipher** if and only if

$$P(M = m \mid C = c) = P(M = m)$$

# Shannon's Theorem

**In a *perfect cipher* the number of keys must be greater or equal to the number of possible messages: $|K| \geq |M|$**

**Sketch of the proof** (by reduction to absurdity):

$C$ must be at least "as large" as $M$: $|C| \geq |M|$.

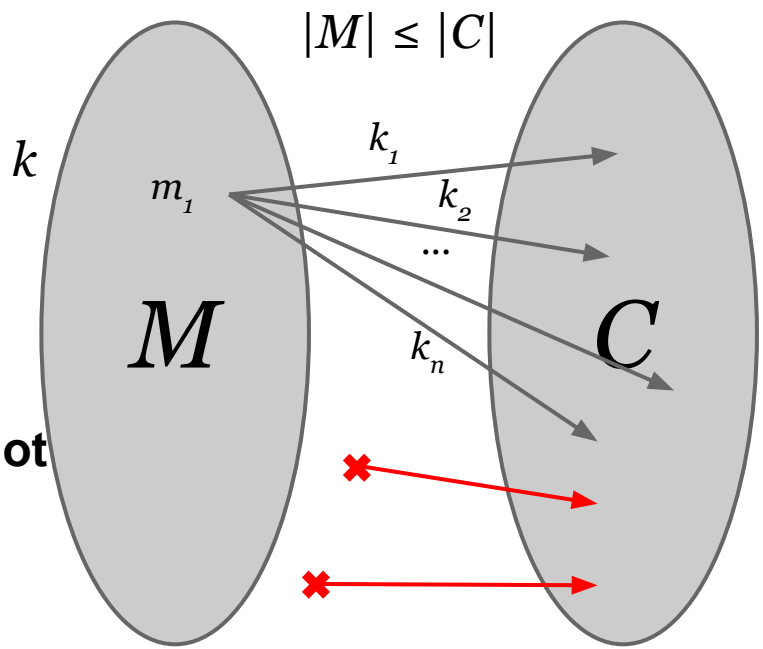
For each *m*, there is one ciphertext *c* for each key $k$

If $|K| < |M|$, the number of ciphertexts for each message would be < than $|M|$. Thus, given that:

$|C| \geq |M| > |K|$

**this means that there are values in $C$ that cannot correspond to (some) of the values in $M$.**

This is **absurd by definition** of perfect cipher.



# The One Time Pad: Perfect Cipher

- XOR of a message $m$ and a random key $k$ of the same size of $m$: $len(k) = len(m)$
  - The key is pre-shared and consumed while writing. Can never be re-used again!
- The OTP is a **minimal perfect cipher**
  - Minimal because $|K| = |M|$
- Terribly inconvenient; used only in special settings
- Real-world algorithms are **not perfect**, and so can be broken

# Imperfections and Brute Force

- Real-world ciphers are imperfect
  - each ciphertext-plaintext pair leaks a small amount of information (because the key is re-used)
- Only thing unknown is the key (Kerckhoffs)
  - Remember: the algorithm itself is known!
- Brute forcing is possible for any <u>real world cipher</u>
  - Try all possible keys, until **one** produces an output that *makes sense.*
  - **<u>Perfect ciphers (one time pads) are not vulnerable</u>** because trying all the (random) keys will yield all the (possible) plaintexts, which are all equally likely (= no clue)

# Cryptanalysis: Breaking Ciphers

A real (non perfect) cryptosystem is **broken** if there is a way to break it that is **faster** than brute forcing.

Types of attacks:

- **Ciphertext attack**: analyst has only ciphertexts
- **Known plaintext attack:** analyst has a set of pairs plain-ciphertext
- **Chosen plaintext attack:** analyst can choose plaintexts and obtain their respective ciphertexts

# Example: can you break this?

- I give you a ZIP-compressed file encrypted with a (secret) 4-bytes key
- I tell you how I encrypted it: algorithm should not be secret (by Kerchoffs):
  - *C* = *K* xor *M*
  - Example:
    - K(hex) = `AA BB CC DD (repeat the key)`
    - M(hex) = `50 4B 03 04 BA DA 55 55 .. .. .. (and so on)`

      XOR
    - C(hex) = `FA F0 CF D9 10 61 99 88 .. .. .. .. .. .. ..`
- I give you a ZIP file encrypted with a key: can you recover the key w/o bruteforcing?

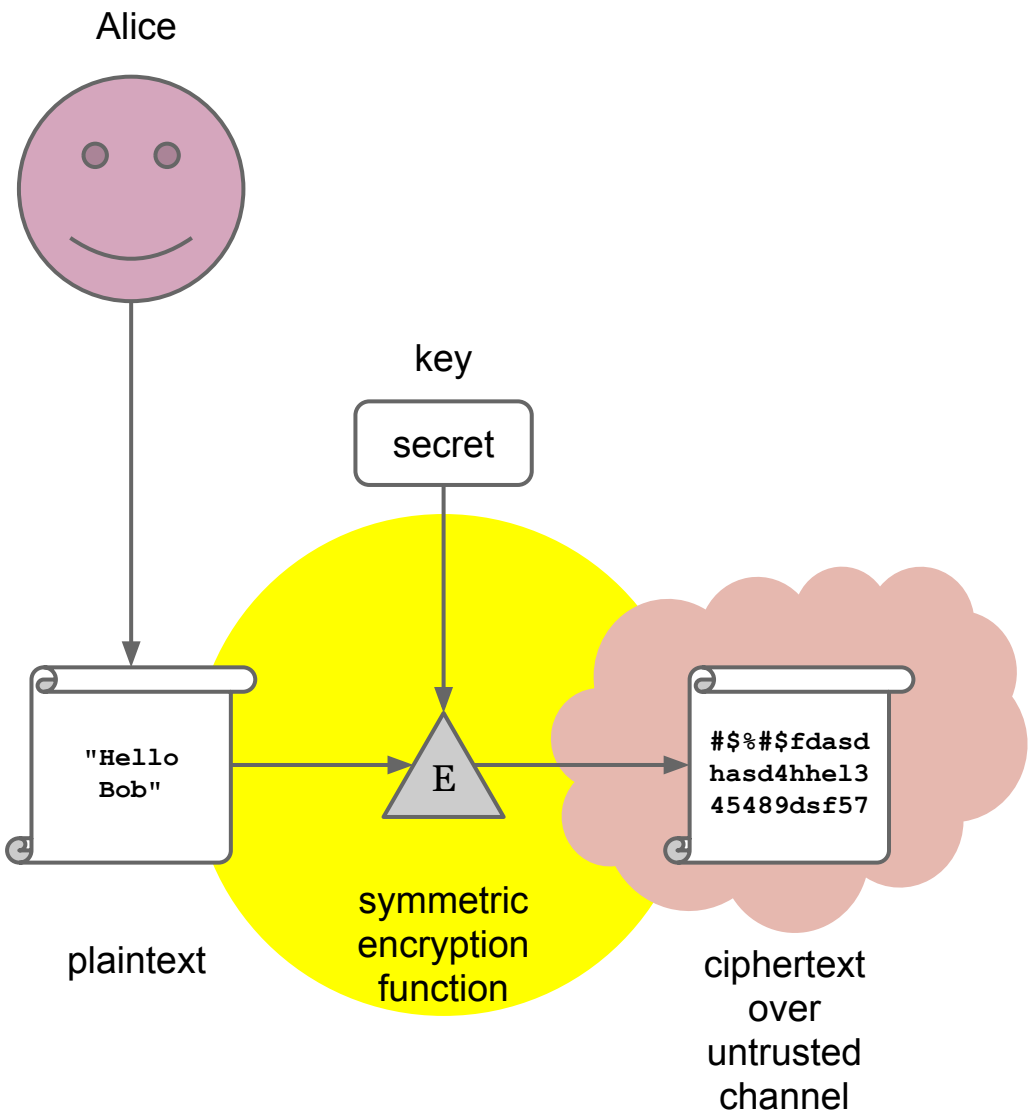
# Key Points to Remember

- Security of a cryptosystem is based on the robustness of the *algorithm.*
- No algorithm, save the one-time-pad, is invulnerable.
- An algorithm is *broken* if there is at least one attack faster than brute forcing.
- There is no way to prove robustness of a cipher, save by trying to break it.
- Secret algorithms are insecure. Security is *transparency*.

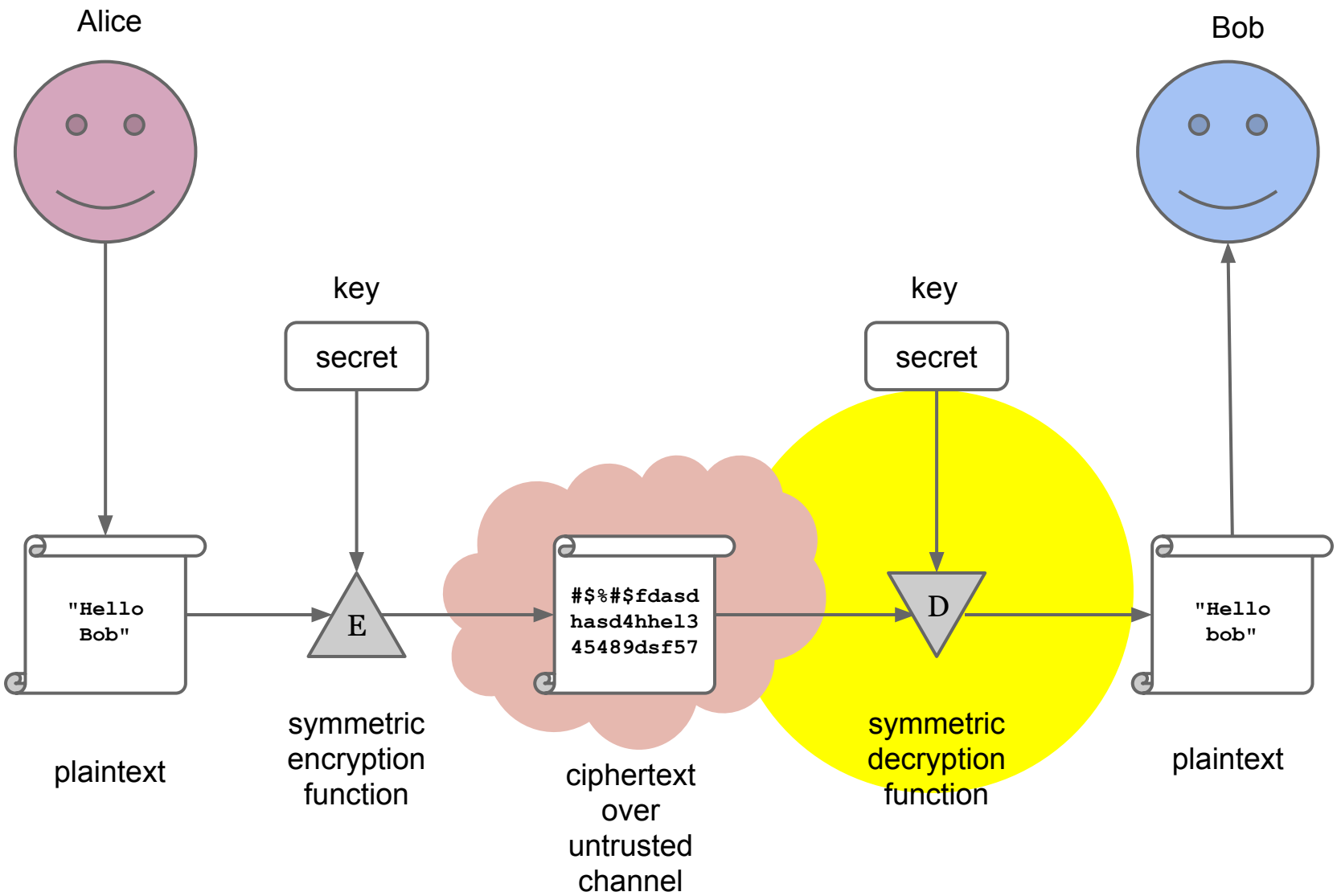
# Symmetric Encryption

Confidentiality

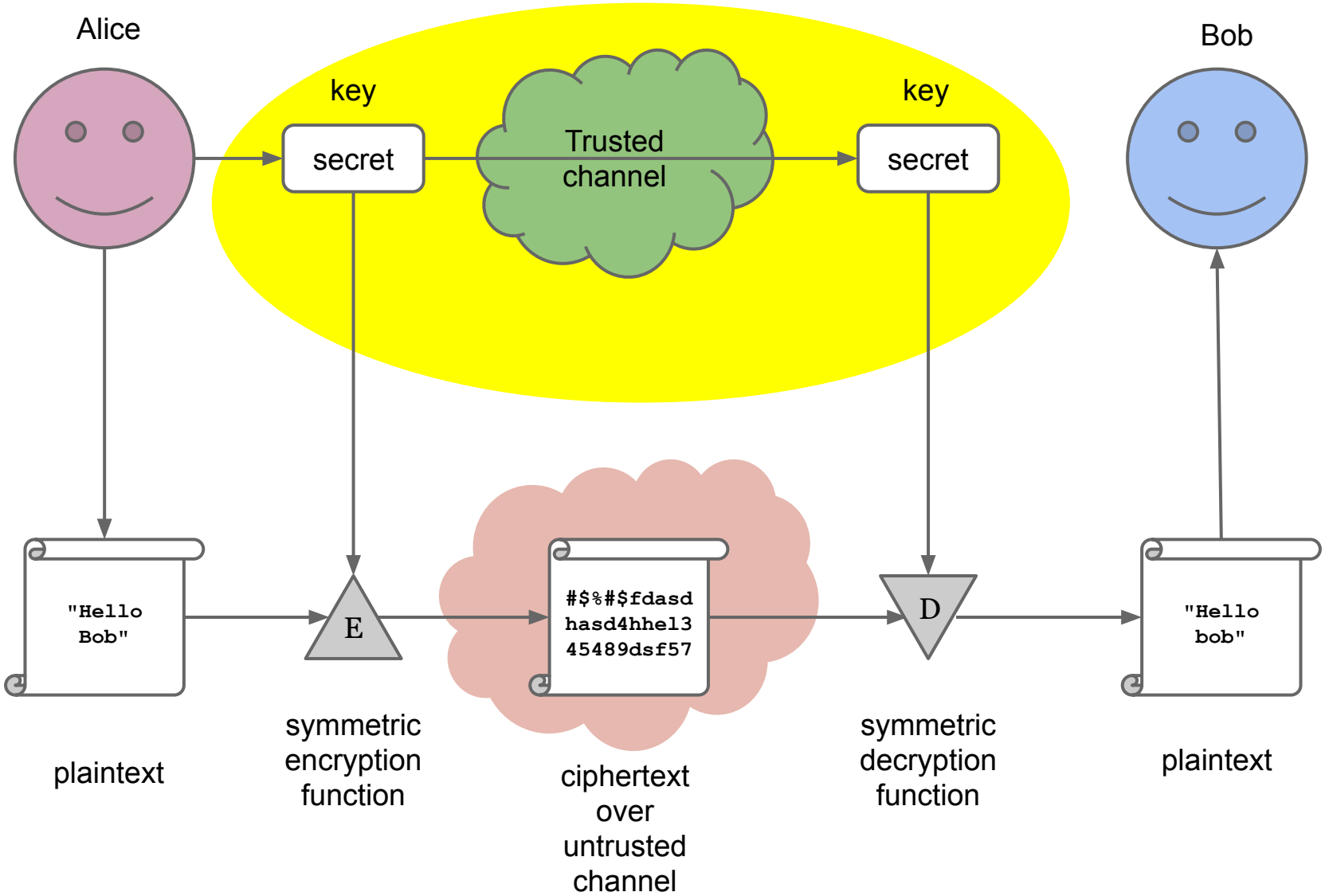
# Symmetric Encryption



# Symmetric Encryption



# Symmetric Encryption



# Symmetric Encryption

- The basic idea of encryption
  - Use key $K$ to ***encrypt*** plaintext in ciphertext
  - Use same key $K$ to ***decrypt*** ciphertext in plaintext
- Synonyms: shared key encryption, secret key encryption
- **Issue:** how do we agree on the key?
  - Cannot send key on same channel as message!
  - Off-band transmission mechanism needed
- **Issue:** scalability
- A symmetric algorithm is a cocktail...

# First ingredient: substitution

**Substitution:** “replacing each byte with another”

Toy example (Caesar cipher)

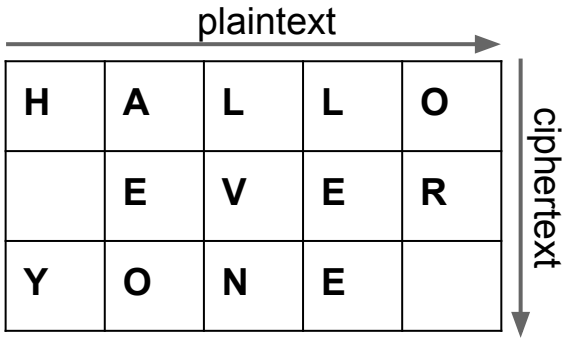
- replace each letter in a sentence with the one following it by $K$ positions in the alphabet
- Example: “**`SECURE`**” becomes “**`VHFXUH`**” with $K = 3$

**Many issues** (it’s a toy example!)**:**

- if cipher known, with 25 attempts at most, 13 on average, we have the key: **keyspace too small**.
- **repetitions** and **structure "**visible" in ciphertext: monoalphabetic ciphers are weak.

# Second ingredient: transposition

Transposition (or diffusion) means “swapping the values of given bits”




Toy example (matrix):

- Write by rows, read by columns
- Key: $K = (R, C)$ with $\mathrm{R} * \mathrm{C} \sim \mathrm{len}(\mathrm{msg})$

**Many issues** (it’s a toy example!):

- Keyspace still relatively small
- But **repetitions** and **structure** gone
- We now really need to test all possible structures

# Modern Symmetric Ciphers

- Modern ciphers mix diffusion and substitution
- Some well known ciphers (just so you recognize the names)
  - DES (Data Encryption Standard, 1977), and its evolution 3DES
  - IDEA (1991)
  - BlowFish (1993)
  - RC5 (1994)
  - CAST-128 (1997)
  - Rijndael (since 2000 it is the AES, Advanced Encryption Standard)

# Case Study: DES

Originally designed by IBM (1973-1974)

Its core function is an S-box (a key-dependent substitution)

It uses a 56 bit key ($2^{56}$ keyspace)

# Case Study: DES vs. NSA

In 1976 it becomes a US standard; its S-boxes are "redesigned" by the NSA

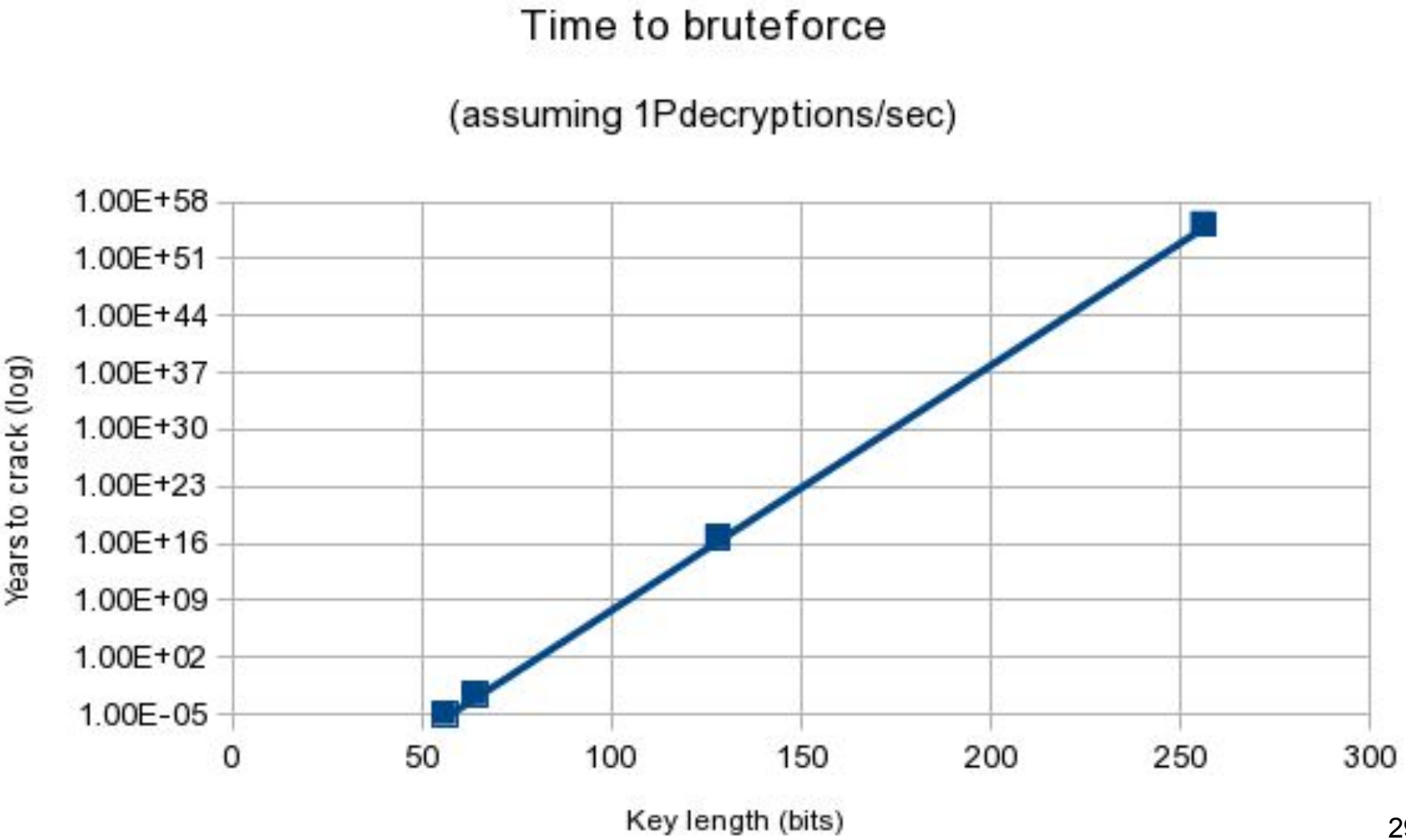
- **Late 1980s:** *differential cryptanalysis* discovered
- **1993**: shown that the original S-boxes would have made DES vulnerable to the differential cryptanalysis, whereas the NSA-designed S-boxes were specifically immune to that.
- **Wait!** Wasn't differential cryptanalysis unknown until late 1980s? *Mmmmmmaybe* the NSA knew about differential crypto in the 70s.

# Keyspace and Brute Forcing

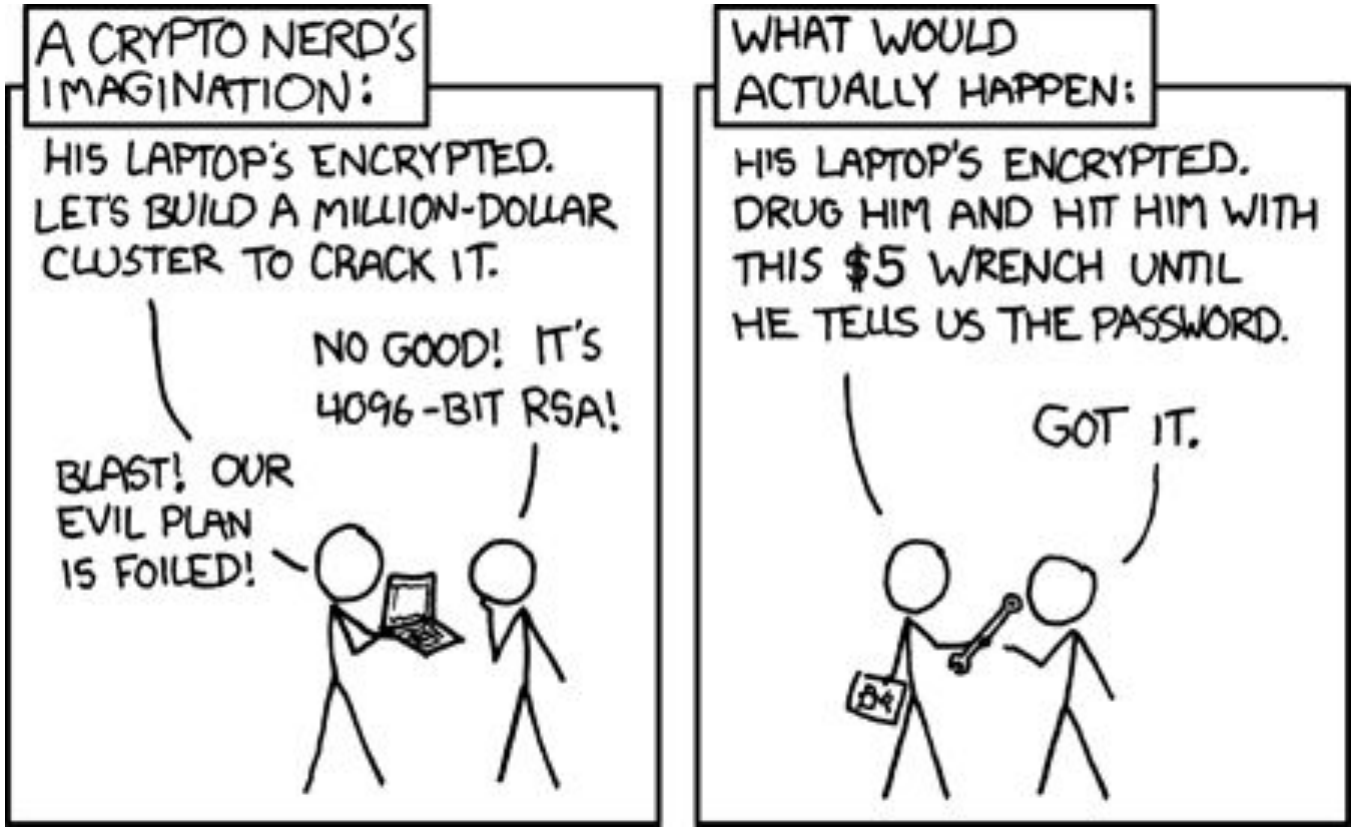
Keyspace generally measured in bits

- Attack time exponential on the number of bits (i.e., 33 bits need twice the time of 32)
- Need to balance computational power vs key length.

# Keyspace *vs.* Time for Brute Forcing



# A good point to remember...



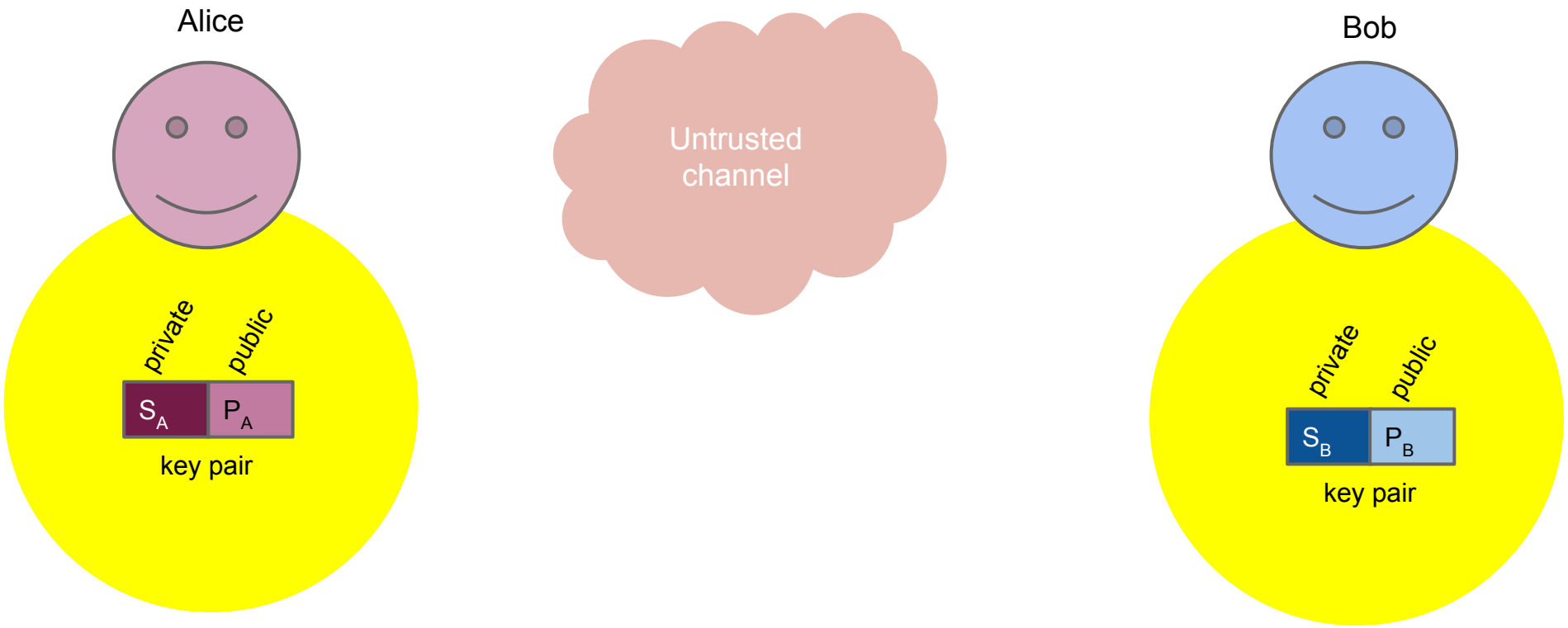
# Asymmetric Encryption

Confidentiality (plus something more)

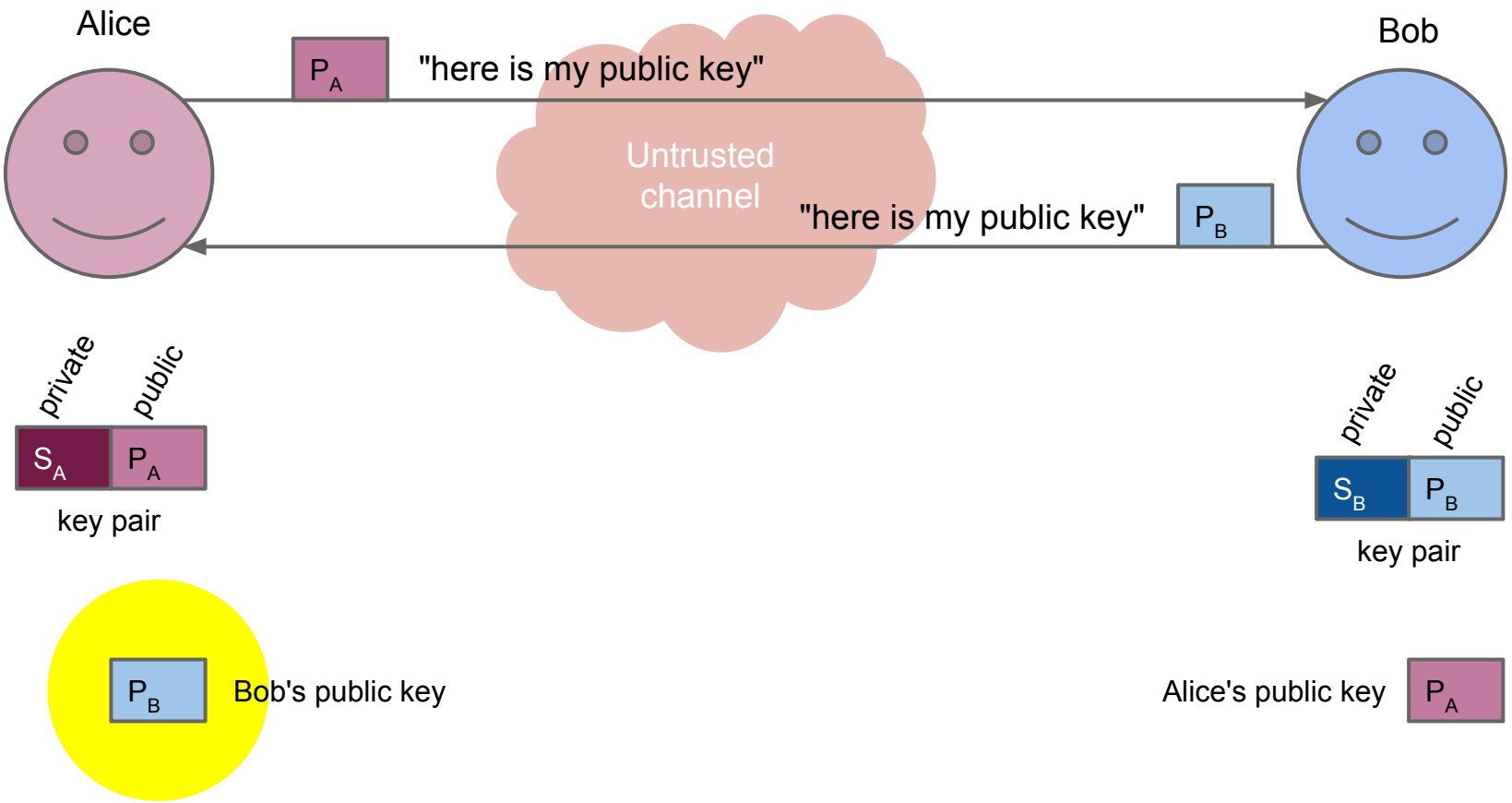
# Asymmetric Encryption

- **Concept:** a cipher that uses two keys
  - What is encrypted with **key1** can be decrypted only with **key2** (and not with key1 itself), and viceversa.
  - The keys cannot be retrieved from each other
- Introduced in 1976 (W. Diffie & M. Hellman)
- Also called *"public key cryptography"*
  - **Idea:** one of the two keys is kept **private** by the subject, and the other can be **publicly** disclosed.
  - This solves radically the problem of **key exchange**
- We will not describe their maths in depth
  - They use a one-way function with a trapdoor
  - They are usually computation-intensive

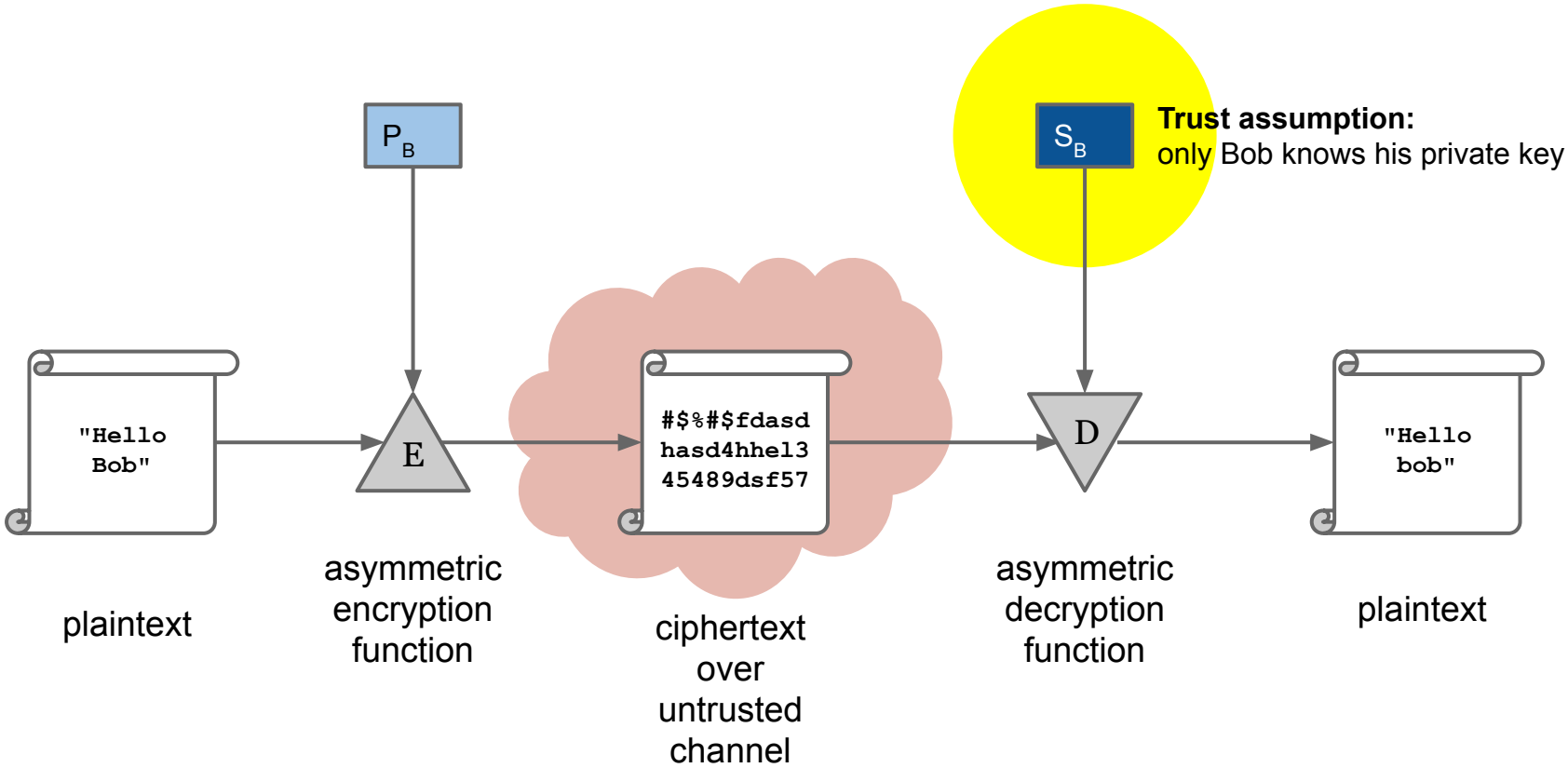
# Asymmetric Encryption: Key Exchange



# Asymmetric Encryption: Key Exchange



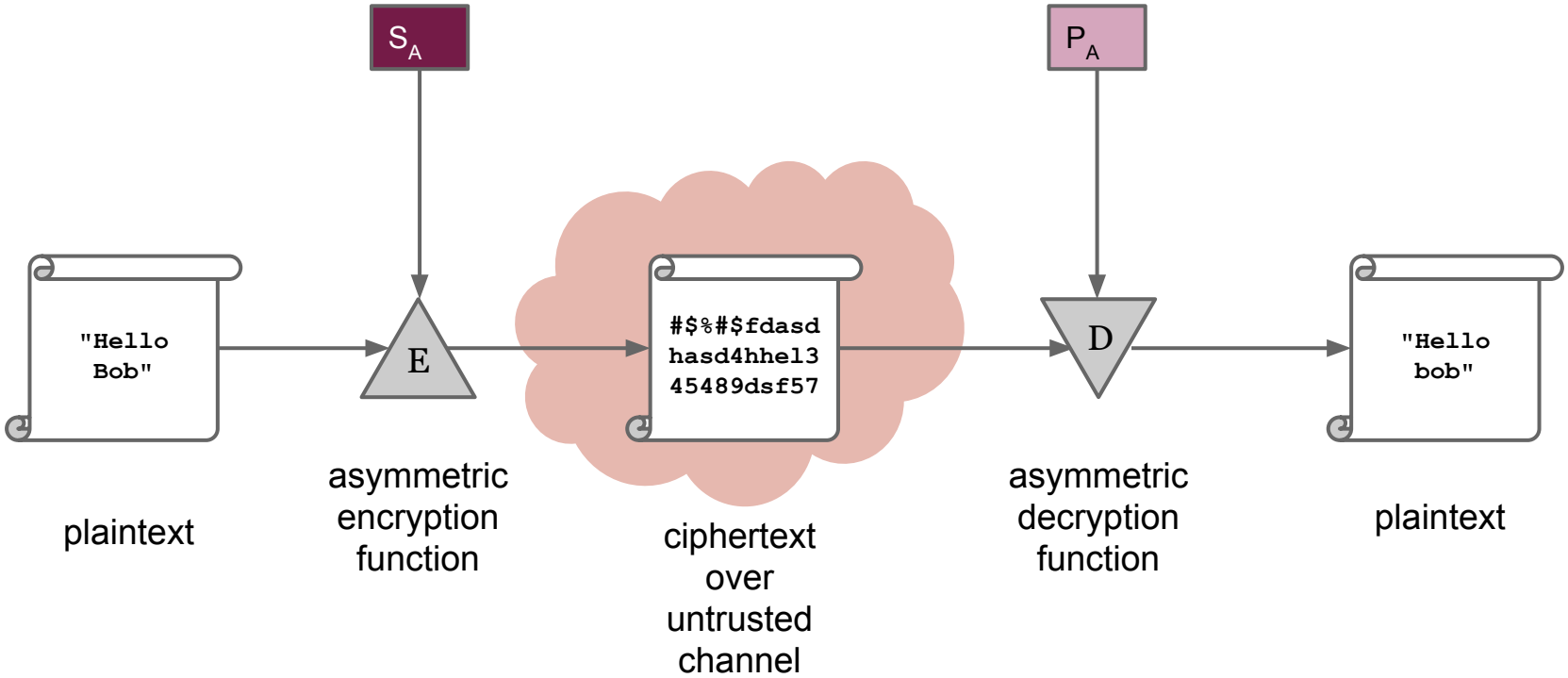
# Asymmetric Encryption: Communication



# Exercise: what is this instead?

**Trust assumption:**
only Alice knows her private key

**Everybody** knows Alice's public key



# Common Asymmetric Ciphers

This is a brief list of names of commonly used asymmetric algorithms:

- Diffie-Hellman (1976)
- RSA (1977, Ron Rivest, Adi Shamir, Len Adleman)
- DSS (1991, FIPS PUB 186)
- ECC (IEEE P1363, elliptic curve cryptography)

# Example: Diffie-Hellman Exchange

- Used by Alice and Bob to agree on a **secret** over an **insecure channel**
  - two people talk in the middle of the classroom, everybody hears them, but at the end only those two people know a secret, and nobody else
- One-way trapdoor: **discrete logarithm**
  - If $y = a^x$ then $x = log_a y$ (Math 101)
  - given $x$, $a$, $p$, it is easy to compute $y = a^x \bmod p$, but knowing $y$, it is difficult to compute $x$
  - Here "difficult" means *"computationally very intensive"*, for all practical purposes the problem requires bruteforce over all possible values of $x$

# How does D-H work (1) - Example

Pick $p$ prime, $a$ primitive root of $p$, public

**Primitive root:** a number $a$ such that raising it to any number between $1$ and $(p - 1)$, $\bmod p$, we obtain each number between $1$ and $(p - 1)$

- Example: $3$ is a primitive root of $7$ because
  - $3^1 \bmod 7 = 3$ $\quad 3^2 \bmod 7 = 2$ $\quad 3^3 \bmod 7 = 6$
  - $3^4 \bmod 7 = 4$ $\quad 3^5 \bmod 7 = 5$ $\quad 3^6 \bmod 7 = 1$

So let $a = 3$, $p = 7$ known to everyone in the system

# How does D-H work (2) - Keys

**Private key** (undisclosed)**:**

They pick a number $X$ in *[1, 2, ..., (p-1)]*

Alice $X_A$

Bob $X_B$

$X_A = 3$
$X_B = 1$

**Public key** (disclosed to everyone):

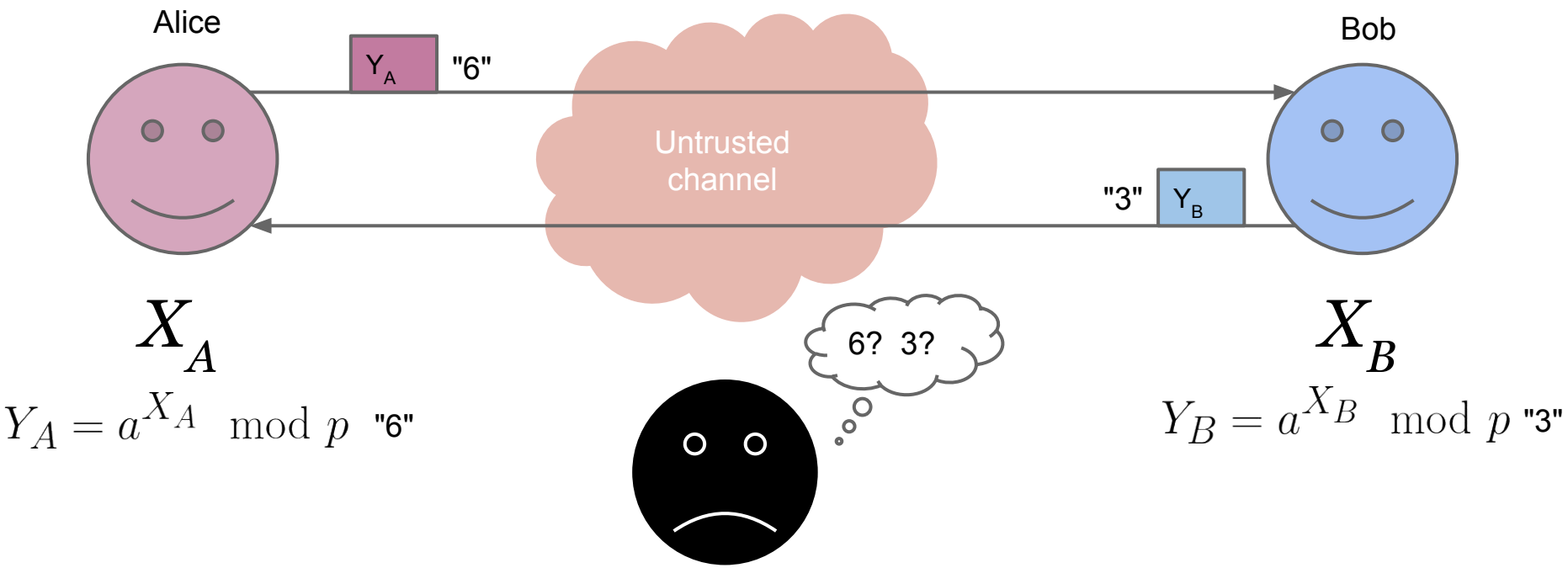
They obtain their public key as

Alice $Y_A = a^{X_A} \mod p$

Bob $Y_B = a^{X_B} \mod p$

$Y_A = 3^3 \bmod 7 = 6$
$Y_B = 3^1 \bmod 7 = 3$

# How does D-H work (3)



# How does D-H work (4) - Secret

At this point, they can compute a ***secret*** $K$

- Since $Y_A^{X_B} \mod p = (a^{X_A} \mod p)^{X_B} \mod p = (a^{X_A})^{X_B} \mod p = (a^{X_B})^{X_A} \mod p = (a^{X_B} \mod p)^{X_A} \mod p = Y_B^{X_A} \mod p$
- Alice $K_A = Y_B^{X_A} \mod p = 3^3 \mod 7 = 6$
- Bob $K_B = Y_A^{X_B} \mod p = 6^1 \mod 7 = 6$

$$\boxed{K_B = K_A = K}$$

Anybody else can listen, but cannot compute $K$

- Because they miss the private key.

# The RSA Algorithm (hints) - 1

Same trick, different base problem (factorization).

If $p$ and $q$ are two ***large primes***:

- computing $n = p * q$ is ***easy***
- but given $n$ it is painfully ***slow*** to get $p$ and $q$
- quadratic sieve field, basically "try all primes until you get to the smaller between $p$ and $q$"
- Different problem than mod-log (D-H), but it can be shown that they are related

# The RSA Algorithm (hints) - 2

- Factoring $n$ is exponential in the number of bits of $n$
- Computation time for encryption grows linearly in the number of bits of *n*
  - square-and-multiply algorithm in hardware
- At the moment of writing:
  - 512-bit RSA factored within 4 hours on EC2 for < $100: http://seclab.upenn.edu/projects/faas/faas.pdf
  - No demonstration of practical factoring of anything larger than 700 bits
  - key sizes > 1024 are safe
  - 2048 or 4096 typical choices

# Key Lengths: caveat

- Key length measured in bits both in **symmetric** and **asymmetric** algorithms
- However, they measure different things
  - Symmetric: number of ***decryption attempts***
  - Asymmetric: number of ***key-breaking attempts***
- Therefore:
  - You can compare symmetric algorithms based on the key (e.g., CAST-128 bit “weaker” than AES-256)
  - You **cannot <u>directly</u> compare** asymmetric algorithms based on key length.
  - More importantly, **never compare directly** asymmetric vs. symmetric key lengths!
  - https://www.keylength.com/en/3/

# The Systems Perspective

“You have probably seen the door to a bank vault…10-inch thick, hardened steel, with large bolts…We often find the digital equivalent of such a vault door installed in a tent. The people standing around it are arguing over how thick the door should be, rather than spending their time looking at the tent.”

(Niels Ferguson & Bruce Schneier, Practical Cryptography)

# Hash Functions

Integrity

# What is a Hash Function

A function $H(\ )$ that maps arbitrary-length input $x$ on fixed-length output, $h$

- **Collisions:** codomain "smaller" than domain.

Computationally infeasible to find:

- input $x$ such that $H(x) = h$ (a specific hash)
  - **preimage attack resistance**
- input $y$ s.t. $y \neq x$ and $H(y) = H(x)$, with a given $x$
  - **second preimage attack resistance**
- couples of inputs $\{x, y\}$ s.t. $H(x) = H(y)$
  - **collision resistance**

# Commonly Used Hash Functions

- SHA-1* (160 bit output) **"2017 Google: collision attack performed (pdf)"**
- SHA-2 / SHA-3 (256)
- MD5 ** (128) **"cryptographically broken and unsuitable for further use"**

```
$ echo "Hello how are you?" | md5                    # MD5
H(x1) = 1961b669810cb300abd9968f4f4f1530

$ echo "Hello how Are you?" | md5                    # MD5
H(x2) = 4e5a7a345bb647e5f4bef0ce37bf7b77

$ echo "Hello how are you?" | shasum                 # SHA-1
H(x1) = a04ff36f4c5499202413539278052 3be88c50049  -

$ echo "Hello how Are you?" | shasum                 # SHA-1
H(x2) = 95e644da66e911faf6ac324a821c14f18f3bb8e7  -
```

# Attacks to Hash Functions (1)

Hash functions may be *broken*.

**1. Arbitrary collision or (1st or 2nd) preimage attack:**

Given a specific hash $h$, the attacker can find

$$x \text{ such that } H(x) = h$$

or, equivalently, given a specific input $x$ can find

$$y \text{ such that } y \neq x \text{ and } H(y) = H(x)$$

<u>faster than brute forcing</u>.

With a $n$-sized hash function, *random* collisions can happen in $(2^{n-1})$ cases.

# Attacks to Hash Functions (2)

**2. Simplified collision attack:**

The attacker can generate colliding couples

$$\{x, y\} \text{ s.t. } H(x) = H(y)$$

<u>faster than brute forcing.</u>

Random collisions can happen in ($2^{n/2}$) cases because of the birthday paradox:

- given *n* randomly chosen people, some pairs will have same birthday
  - probability = 100% if ***n* = 367**
  - probability = 99.9% if ***n* = 70** people
  - probability = 50% if *n* = 23 people, and so on...
- vs. very low chances that some of you are born on a specific date

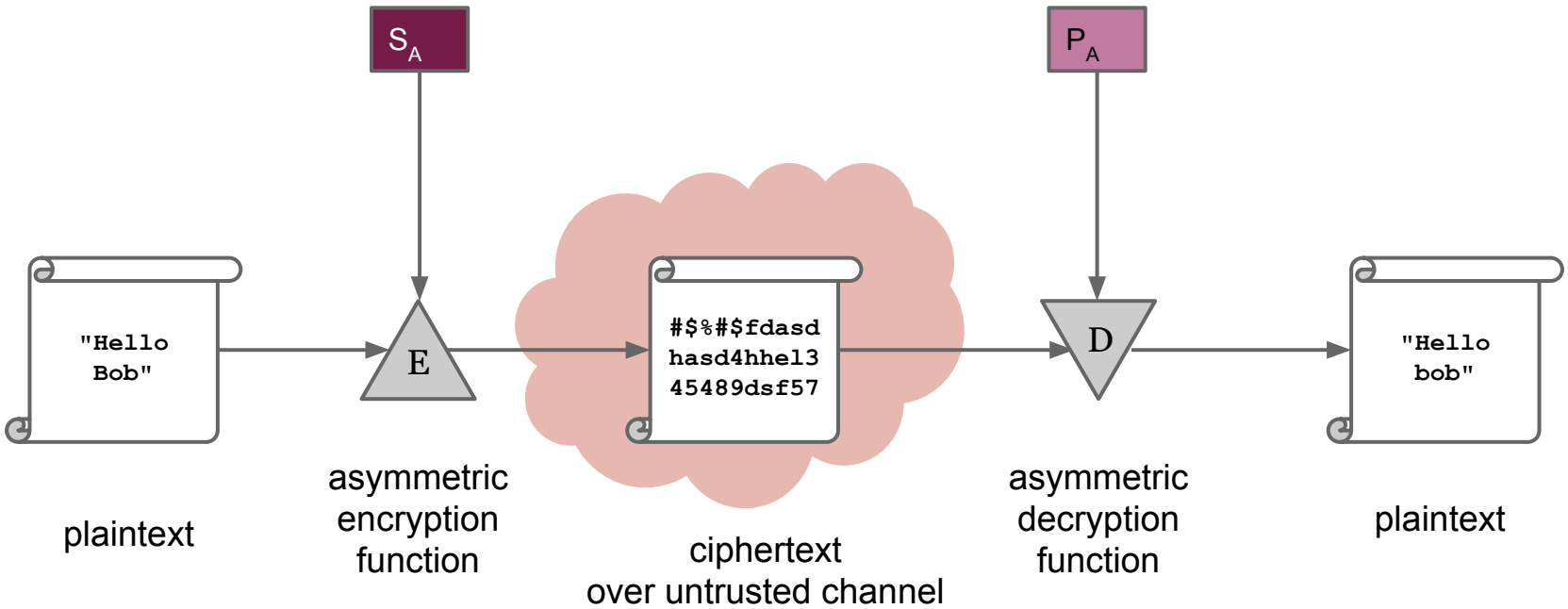
# Digital Signature

Confidentiality, Integrity and Authentication
*(not to be confused with Availability)*

# Digital Signature: Message <u>Authentication</u>

**Trust assumption:**
only Alice knows her private key

**Everybody** knows Alice's public key



# Digital signature: Authentication and Integrity

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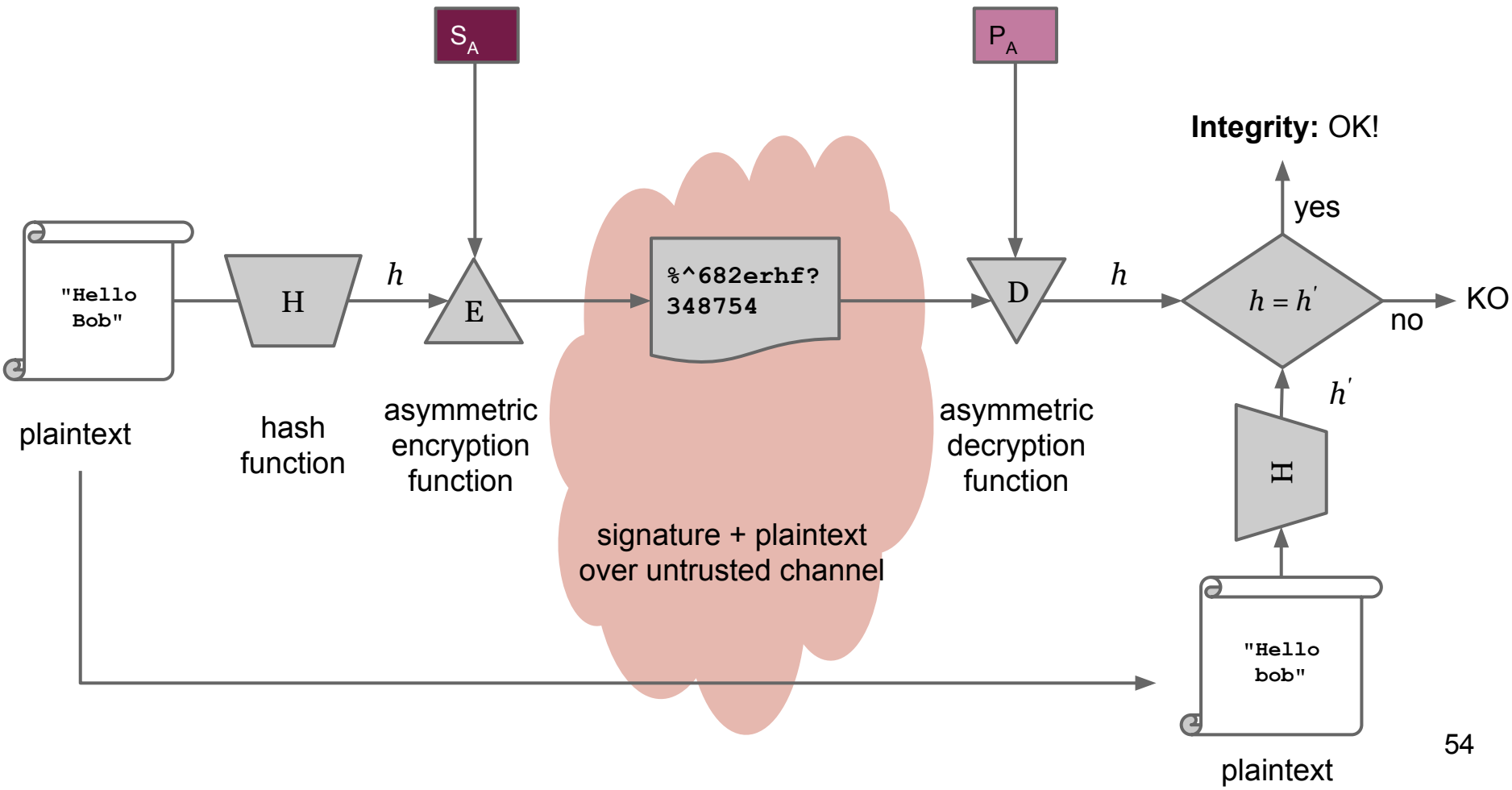



**Integrity:** OK!

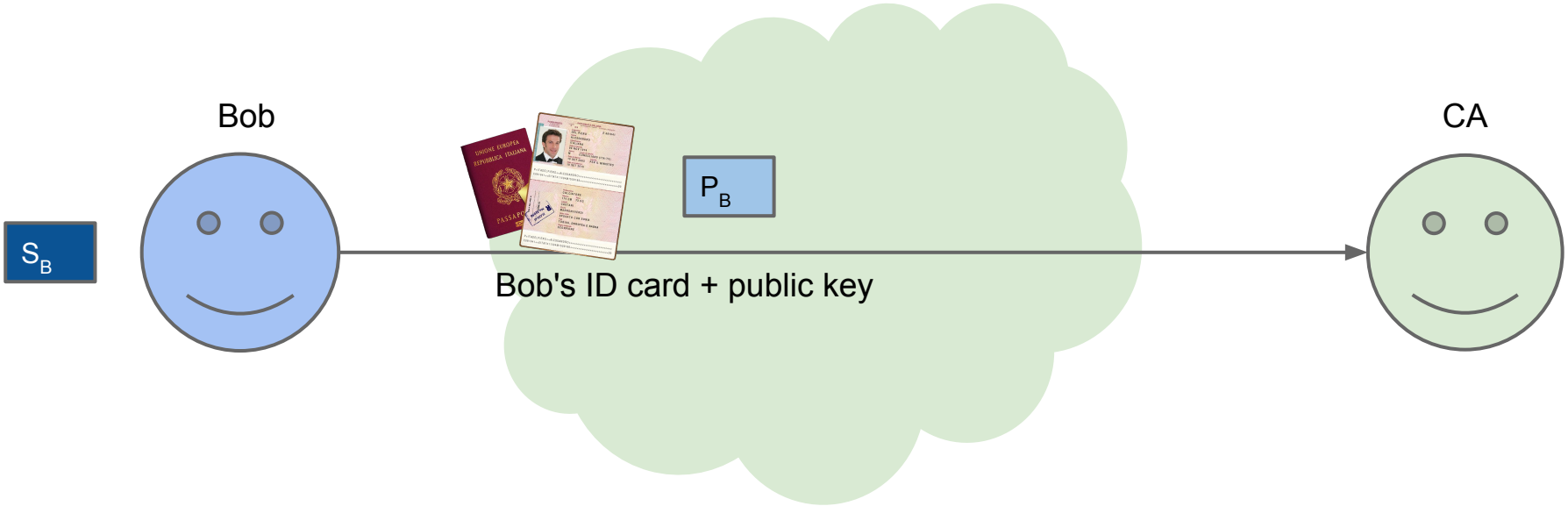
# A case of identity

- A digital signature ensures that plaintext was authored by someone.
- **Not really!** It ensures it was encrypted with a certain key...it says nothing about “who” is using that private key
- Ditto for using public key for encryption!
- Exchange of public keys **must be secured** (either out of band, or otherwise)
- PKI (Public Key Infrastructure) associates keys with identity on a wide scale

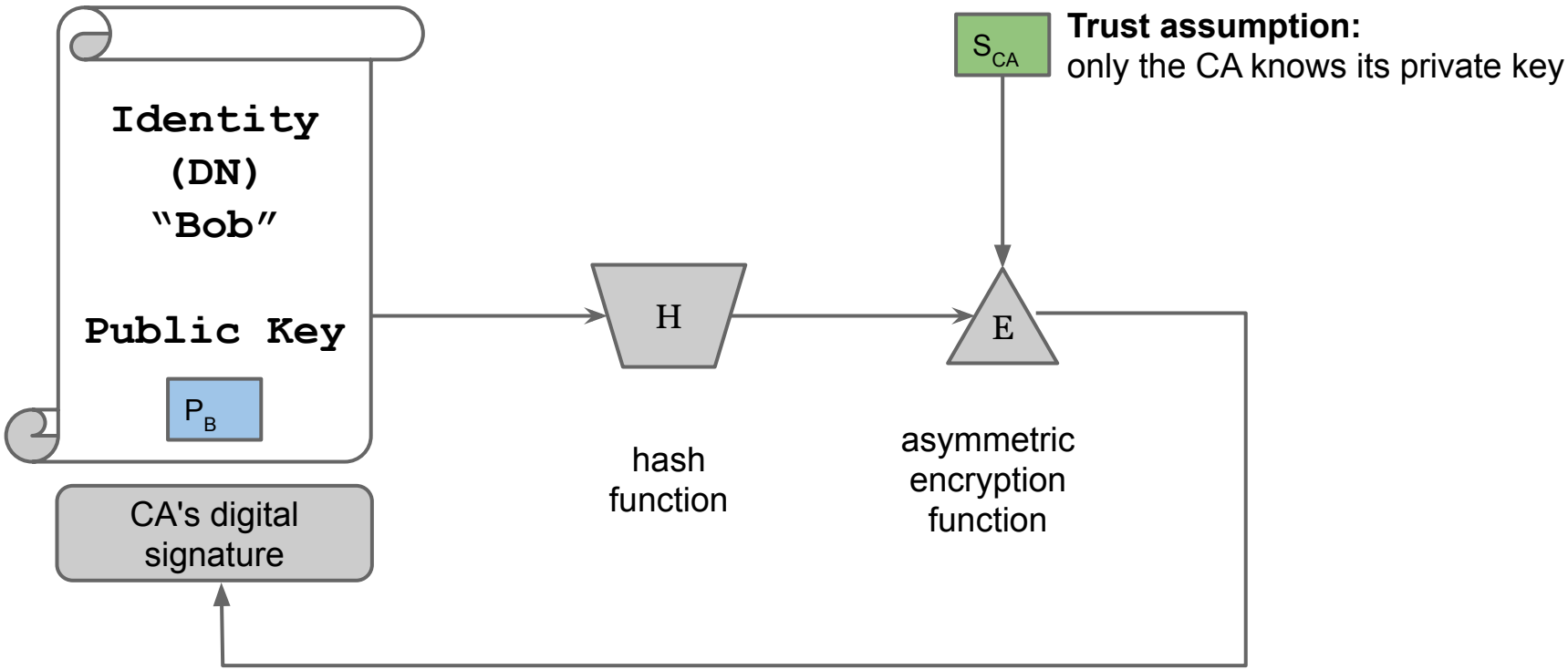
# PKI

- A PKI uses a trusted third party
  - Called a **certification authority** (CA)
- The CA **digitally signs** files called **digital certificates**, which bind an identity to a public key
  - Identity = “Distinguished Name (DN)”
  - As defined in the X.509 standard (most used one)
- Now we can recognize a number of subjects...provided that we can obtain the **public key** of the CA

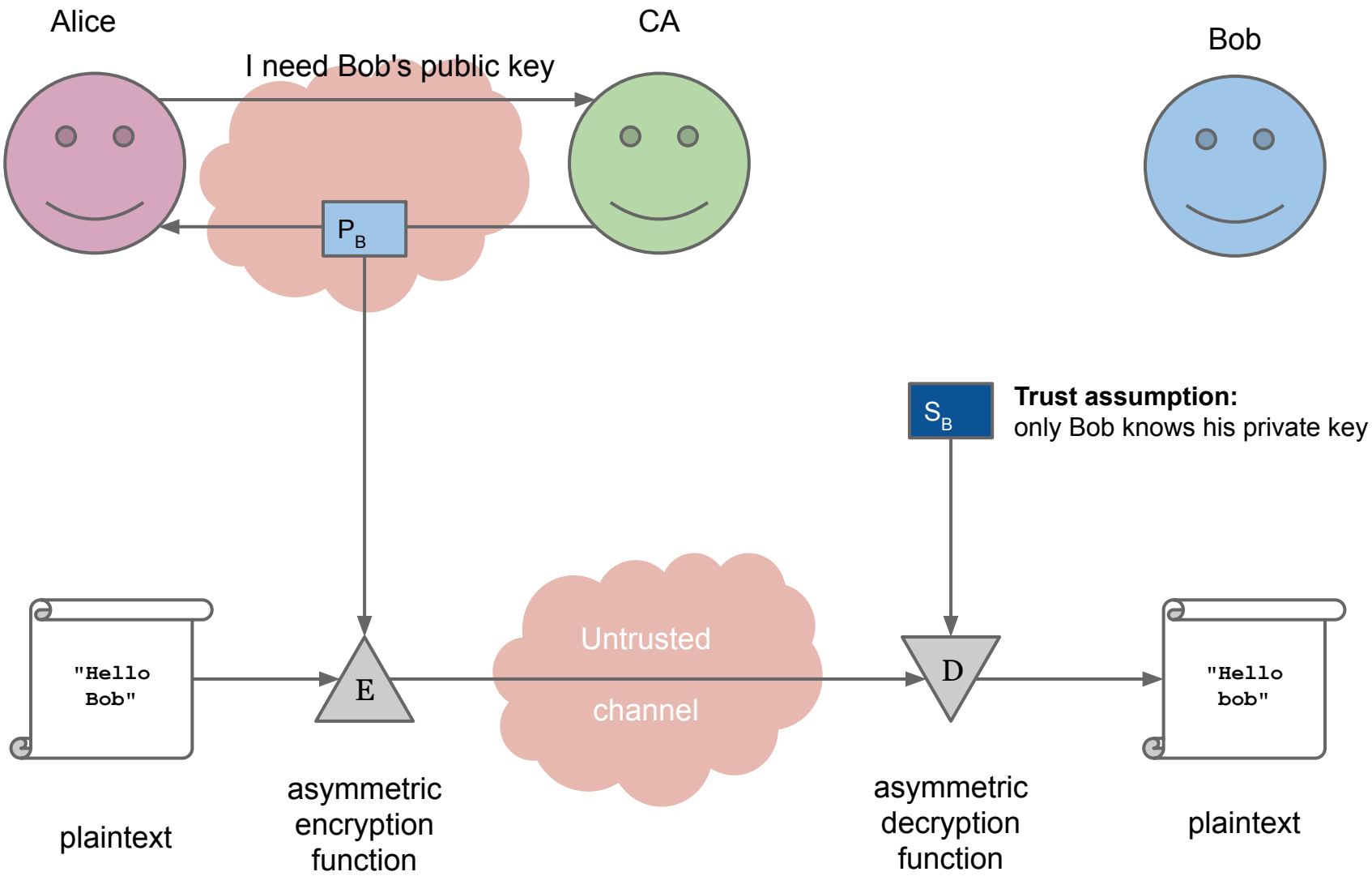
# Bob's Digital Certificate (1)



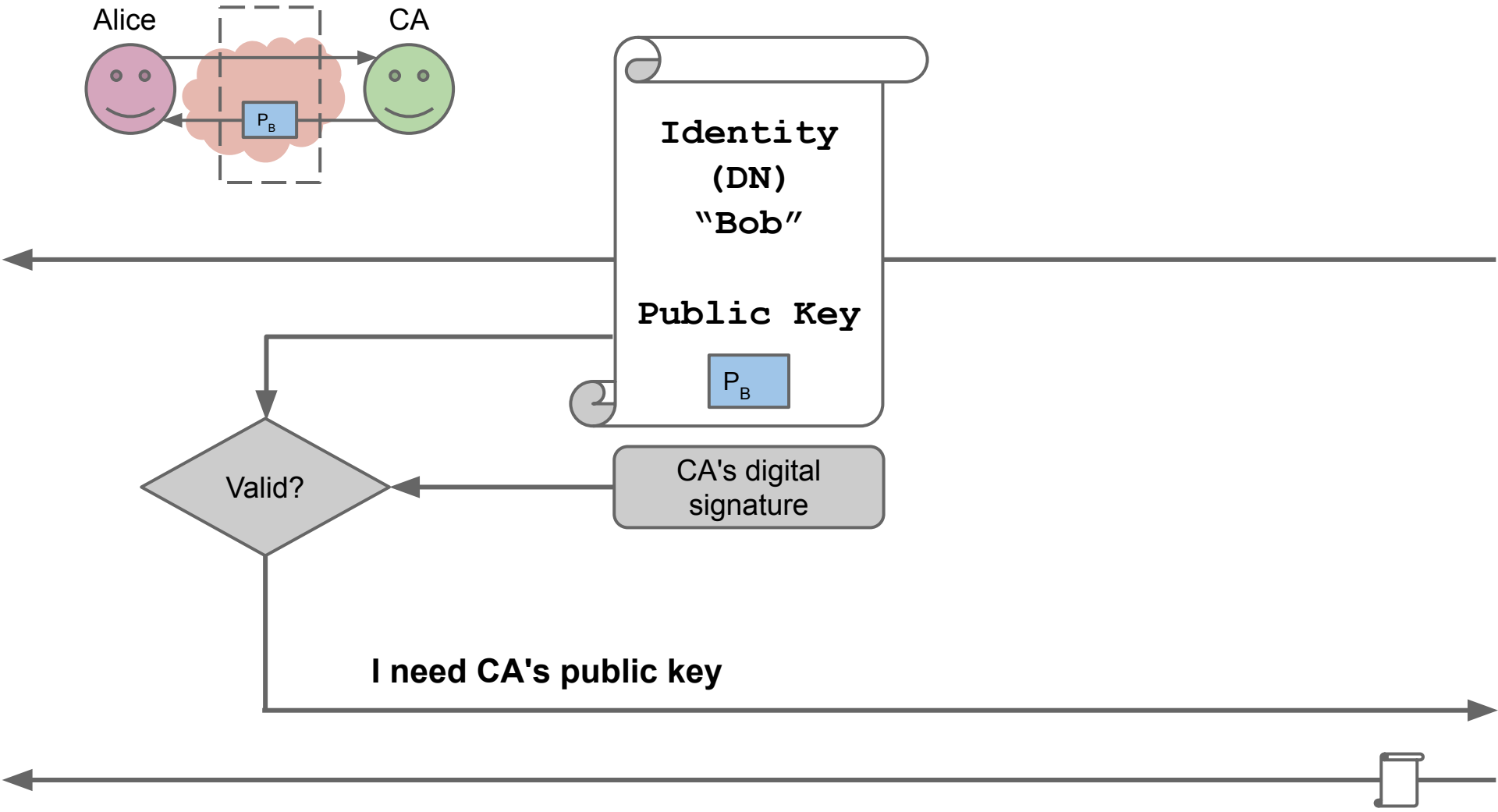
# Bob's Digital Certificate (2)



# Retrieving Bob's Certificate



# Zoom in: Is the public key valid?




**Where is this going to end?**

# The Certificate Chain

***"Quis custodiet custodes?"***

The CA needs a *private key* to sign a certificate

- The *public key*...must be in a certificate.

Someone else needs to sign **that** certificate

- And so on…at some point this needs to stop!

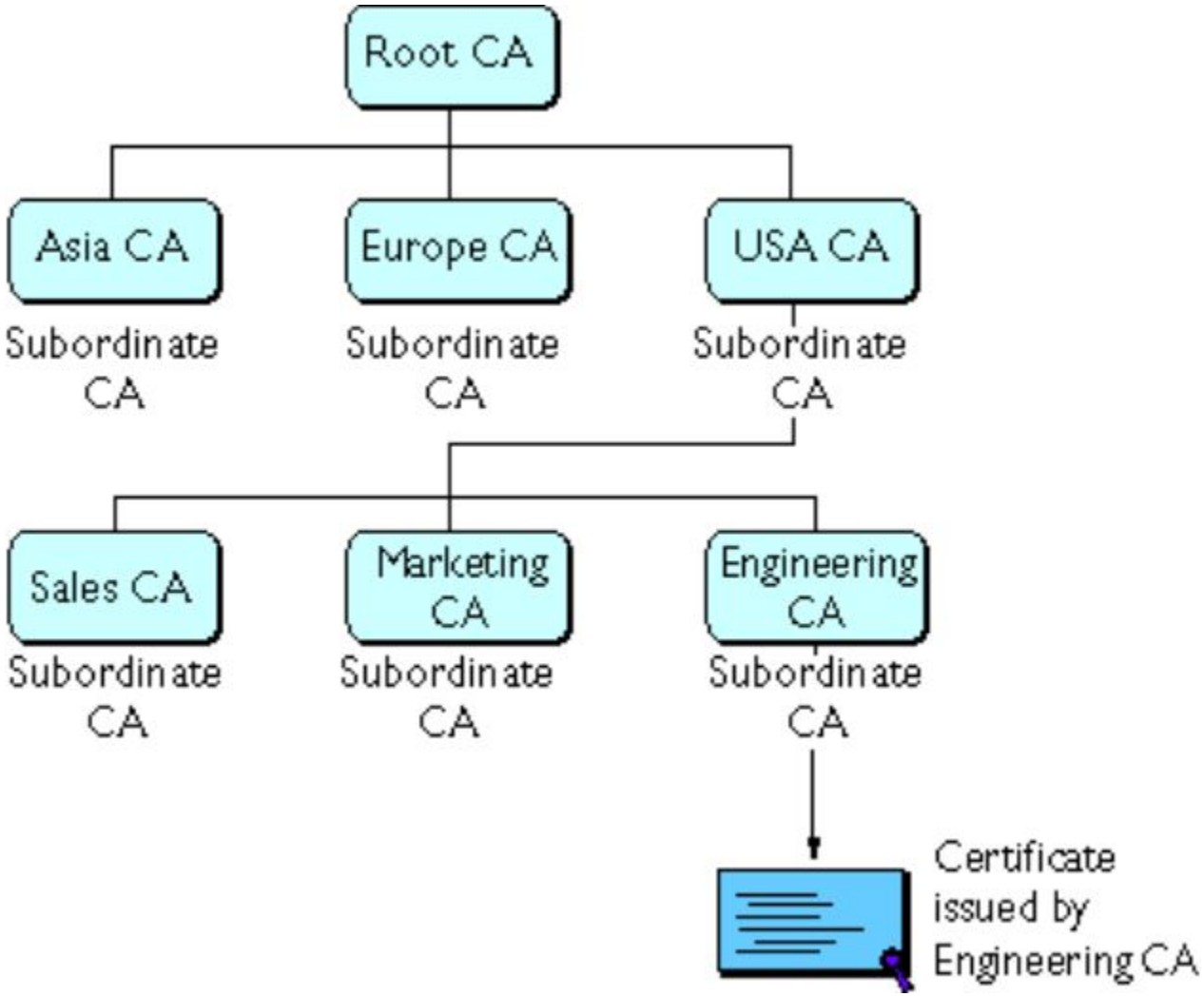
# We Need a Trusted Element

**Top-level CA** (root CA, source CA)

Uses a self-signed certificate

- cannot be verified: it's a **trusted element**
- Basically a document that says “I am myself”

# The chain, or rather the tree



# How to distribute the trusted element?

An **authority** releases it

- the state
- a regulator
- the organization management

Decentralizing trust (e.g., PGP web-of-trust)

CA already (de facto standard)

Keychain Access

Click to unlock the System Roots keychain.

**Keychains**

- login
- PrivateEncryptedDatak
- PrivateEncryptedDatak
- Local Items
- System
- System Roots

**Category**

- All Items
- Passwords
- Secure Notes
- My Certificates
- Keys
- Certificates

**VeriSign Universal Root Certification Authority**

Root certificate authority

Expires: Wednesday 2 December 2037 00 h 59 min 59 s Central European Standard Time

This certificate is valid

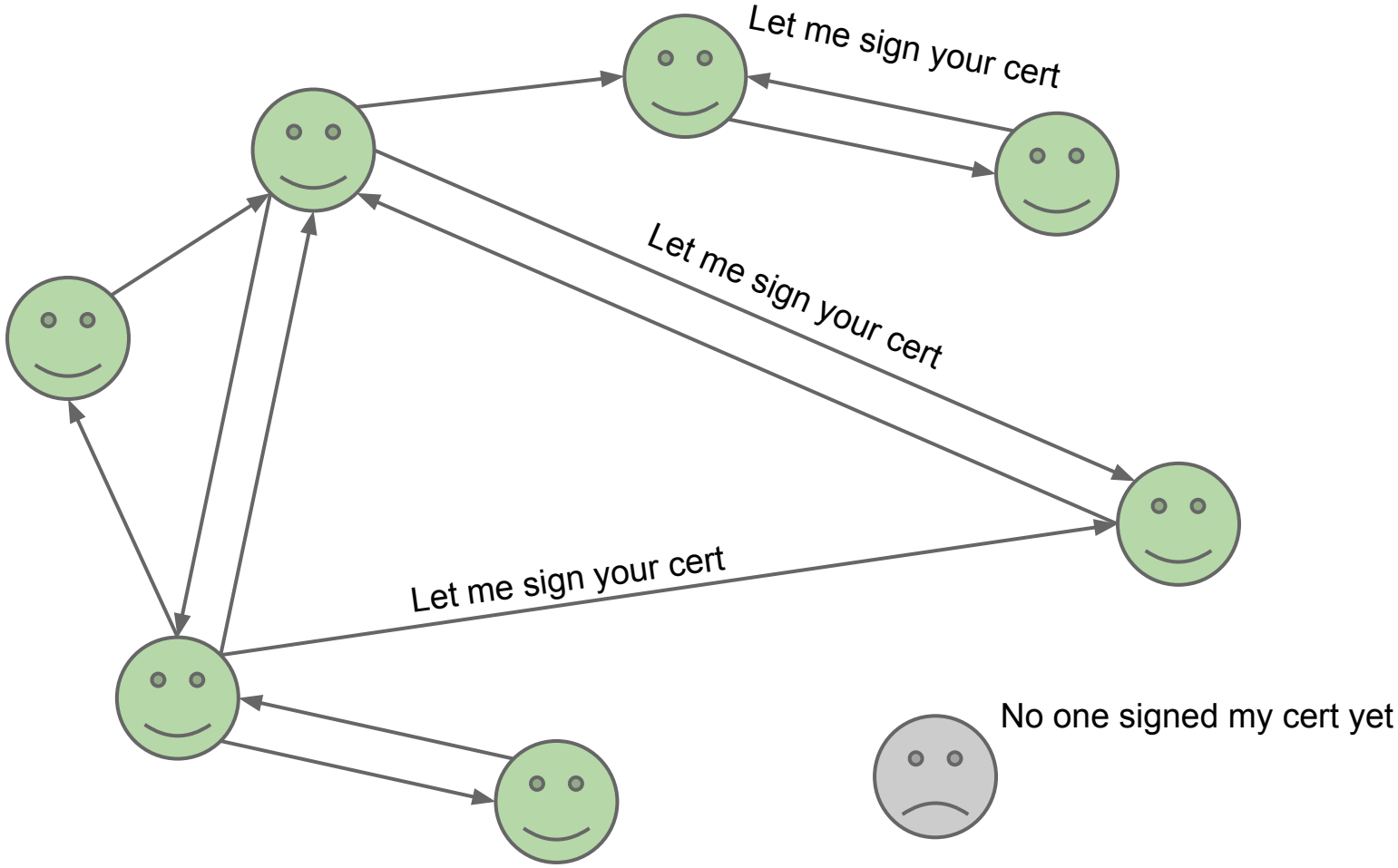
| Name | Kind | Expires | Keychain |
|---|---|---|---|
| XRamp Glob...ification Authority | certificate | 01 Jan 2035 06:37:19 | System Roots |
| WellsSecure...ertificate Authority | certificate | 14 Dec 2022 01:07:54 | System Roots |
| VRK Gov. Root CA | certificate | 18 Dec 2023 14:51:08 | System Roots |
| Visa Informa...Delivery Root CA | certificate | 29 Jun 2025 19:42:42 | System Roots |
| Visa eCommerce Root | certificate | 24 Jun 2022 02:16:12 | System Roots |
| VeriSign Univ...fication Authority | certificate | 02 Dec 2037 00:59:59 | System Roots |
| VeriSign Clas...on Authority - G3 | certificate | 17 Jul 2036 01:59:59 | System Roots |
| VeriSign Clas...on Authority - G5 | certificate | 17 Jul 2036 01:59:59 | System Roots |
| VeriSign Clas...on Authority - G4 | certificate | 19 Jan 2038 00:59:59 | System Roots |
| VeriSign Clas...on Authority - G3 | certificate | 17 Jul 2036 01:59:59 | System Roots |
| VeriSign Clas...on Authority - G3 | certificate | 17 Jul 2036 01:59:59 | System Roots |
| VeriSign Clas...on Authority - G3 | certificate | 17 Jul 2036 01:59:59 | System Roots |
| VAS Latvijas Pasts SSI(RCA) | certificate | 13 Sep 2024 11:27:57 | System Roots |
| UTN-USERFirst-Object | certificate | 09 Jul 2019 20:40:36 | System Roots |
| UTN-USERFir...work Applications | certificate | 09 Jul 2019 20:57:49 | System Roots |
| UTN-USERFirst-Hardware | certificate | 09 Jul 2019 20:19:22 | System Roots |
| UTN-USERFir...tication and Email | certificate | 09 Jul 2019 19:36:58 | System Roots |
| UTN - DATACorp SGC | certificate | 24 Jun 2019 21:06:30 | System Roots |
| UCA Root | certificate | 31 Dec 2029 01:00:00 | System Roots |
| UCA Global Root | certificate | 31 Dec 2037 01:00:00 | System Roots |
| TWCA Root C...fication Authority | certificate | 31 Dec 2030 16:59:59 | System Roots |
| TWCA Global Root CA | certificate | 31 Dec 2030 16:59:59 | System Roots |
| TÜRKTRUST...Hizmet Sağlayıcısı | certificate | 22 Mar 2015 11:27:17 | System Roots |
| TÜRKTRUST...Hizmet Sağlayıcısı | certificate | 16 Sep 2015 12:07:57 | System Roots |
| TÜRKTRUST...Hizmet Sağlayıcısı | certificate | 22 Dec 2017 19:37:19 | System Roots |

Copy

222 items

Do you trust your operating system? Do you trust the list of root certificates that ship with it?

# Decentralizing Trust: Web of Trust



# Certificate Revocation Issues

- Signatures cannot be revoked (destroyed).
- **Certificates** need to be revoked at times.
- Certificate Revocation Lists (CRL)

# Verification Sequence for Certificates

1. Does the **signature** validate the document?
   - Hash verification as we have seen
2. Is the **public key** the one on the certificate?
3. Is the certificate the one of the **subject**?
   - Problems with omonimous subjects, DN
4. Is the **certificate** validated by the CA?
   - Validate the entire certification chain, up to the root
5. Is the **root** certificate trusted?
   - Need to be already in possession of the root cert
6. Is the certificate in a **CRL**?
   - How do we get to the CRL if we are not **online**?

**Any missing check = vulnerability!**

# Case study: Italian “legal” digital signatures framework

Introduced in Italy with D.P.R. 513/97

- many modifications, in particular when implementing EU regulations

Original Italian scheme: a list of “screened” CAs

**Result:** each CA created their own **digital signature application** (i.e., trusted element)

# Attacking Digital Signature Applications

Digital signature stronger than handwritten signature

- **Written** documents can be modified, written signatures can be copied.
- **Digital** signature value **tied to content**, and cannot be forged unless the algorithm is broken
- However, a digital signature is **brittle**: if a fake is forged, it cannot be told from real one

# Crypto: *OK* – Software Design: *KO*

Italian signature standards use **strong**, **unbroken cryptographic algorithms**!

However, vulnerabilities did emerge

Do you remember the “bank vault door in a tent”?

# Bug 1: Fields of pain

- Bug notified on 9/9/2002
- The software of several CAs (originally DiKe by Infocamere was the subject of scrutiny) allowed users to sign Word documents with **dynamic fields** or **macros** without notice
- A macro does not change the bit sequence of the document, so the **signature does not change** with the visualized content
- Examples and stuff on Prof. Zanero's home: http://home.deib.polimi.it/zanero/bug-firma.html (Italian only, sorry for that!)

# Example

provaDiKe.doc - Microsoft Word

File Edit View Insert Format Tools Table Window Help

Normal Arial 11

Questo è un documento di prova scritto con MS Word 2000.
Nelle righe seguenti sono inseriti campi variabili tra asterischi:

Data e ora *17/09/2002 08.48.53*
I dati inseriti sono 17/09/2002 08.48.53

Data e ora *17/09/2002 08.48.53* (Questa non dovrebbe modificarsi)
I dati inseriti sono 17/09/2002 08.48.53

Nome del documento *provaDiKe.doc*
Dato inserito provaDiKe.doc

Nome autore *Manlio Cammarata*
Dato inserito Manlio Cammarata

Nome utente *Manlio Cammarata*
Dato inserito Manlio Cammarata

Ora il file sarà firmato digitalmente con DiKe e dovremo controllare se la verifica va a buon fine anche dopo l'aggiornamento dei campi.

Page 1 Sec 1 1/1 At 2.5cm Ln 1 Col 1 REC TRK EXT OVR Italian (Italy

DiKe

File Modifica Strumenti Opzioni Guida

Nr. Firmatari: 1
Firmatario:1 CAMMARATA MANLIO;Cod. Fiscale:CMMMNL47P04L424H;Identificativo:2001111111812
Stato:IT;Organizzazione:Non Dichiarato;Unità Organizzativa:RA=INFOCAMERE S.C.p.A.
verifica completata correttamente

**Documento: d:\test\provaDiKe18.doc.p7m Dimensioni: 21944 Data: 18/09/2002 14.20.44**

Questo è un documento di prova scritto con MS Word 2000.
Nelle righe seguenti sono inseriti campi variabili tra asterischi:

Data e ora *18/09/2002 14.28.15*
I dati inseriti sono 17/09/2002 08.48.53

Data e ora *17/09/2002 08.48.53* (Questa non dovrebbe modificarsi)
I dati inseriti sono 17/09/2002 08.48.53

Nome del documento *provaDiKe.doc*
Dato inserito provaDiKe.doc

Nome autore *Manlio Cammarata*
Dato inserito Manlio Cammarata

Nome utente *Manlio Cammarata*
Dato inserito Manlio Cammarata

Ora il file sarà firmato digitalmente con DiKe e dovremo controllare se la verifica va a buon fine anche dopo l'aggiornamento dei campi.

# Reactions

- The CAs responded that this was *“intended behavior”* and that it did not violate the law
- However:
  - Microsoft, on 30/1/03, released an Office patch to allow **disabling macros** via API.
  - Nowadays, all software show a big alert when signing an Office document.
  - New legislation explicitly excludes modifiable and scriptable formats (but recommends PDF)
- The issue is actually much deeper
  - Decoders of complex formats should also be validated
  - Research field of “what you see is what you sign”

# Bug 2: Firma&Cifra

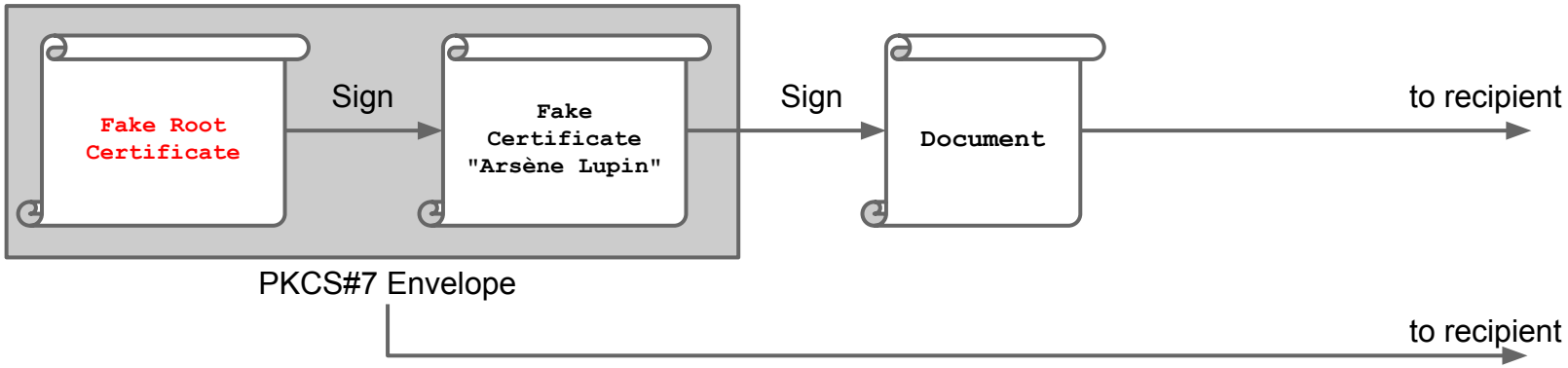
- Firma&Cifra was the digital signature application by PostECom
- Bug found by anonymous on 20/03/2003: http://www.interlex.it/docdigit/sikur159.htm
- **Result: creation** and **verification** of a **signature** with a **fake certificate**
- Also in this case: no cryptographic algorithm was broken to perform the show

# Vulnerability Description

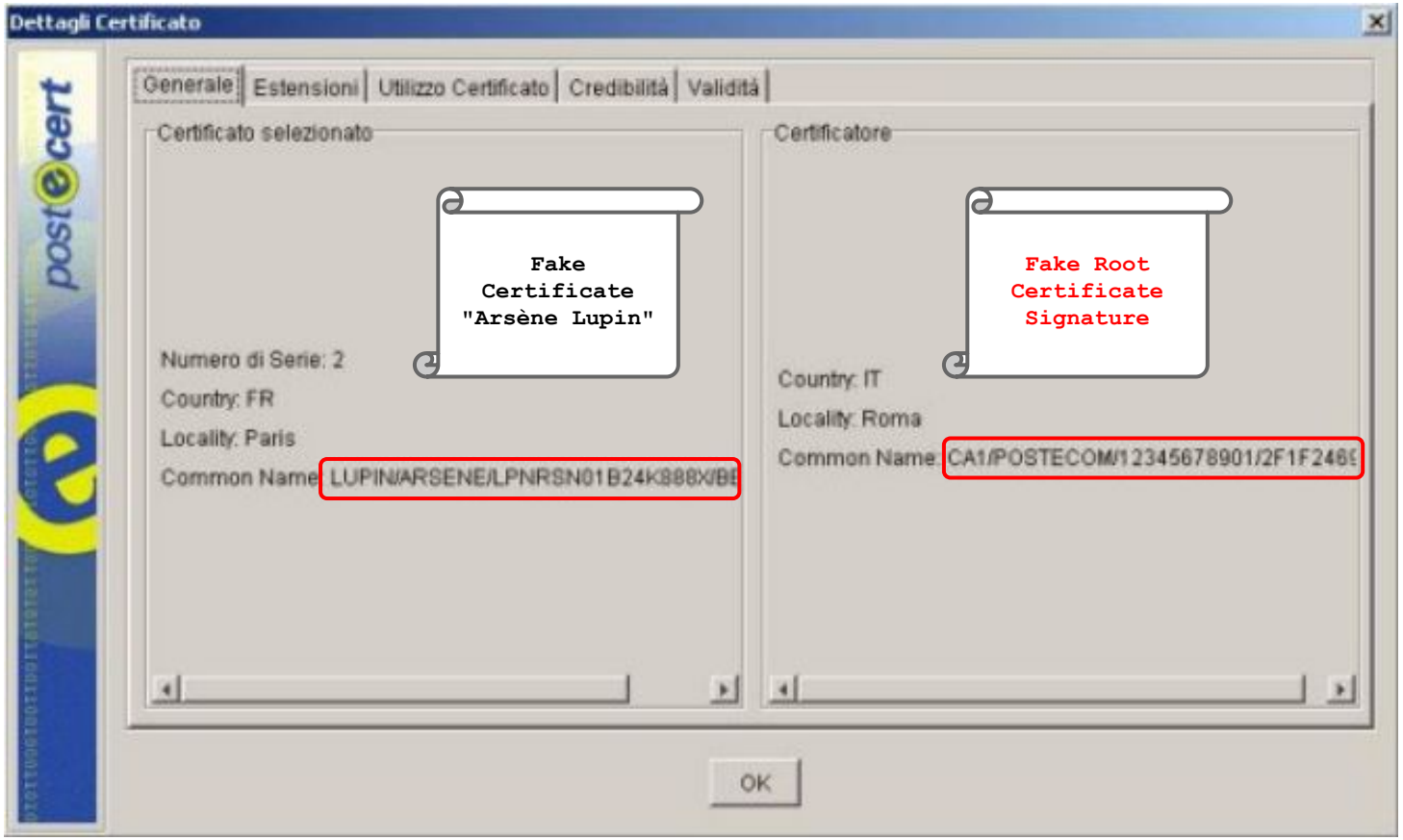
- In order to verify a signature, we need **author certificate** and the **certificate chain**
  - **Theoretically**, all available **online**
  - To allow **offline verification**, everything included with the document, in a PKCS#7 envelope
- Verification of root certificates must use preinstalled ones
  - Most **software** comes with them
  - The **root certificate storage** is a critical point!
- Firma&Cifra **trusts the root certificate in the PKCS#7 envelope**, and it even imports it in the secure storage area.

# The Exploit: Arséne Lupin signature

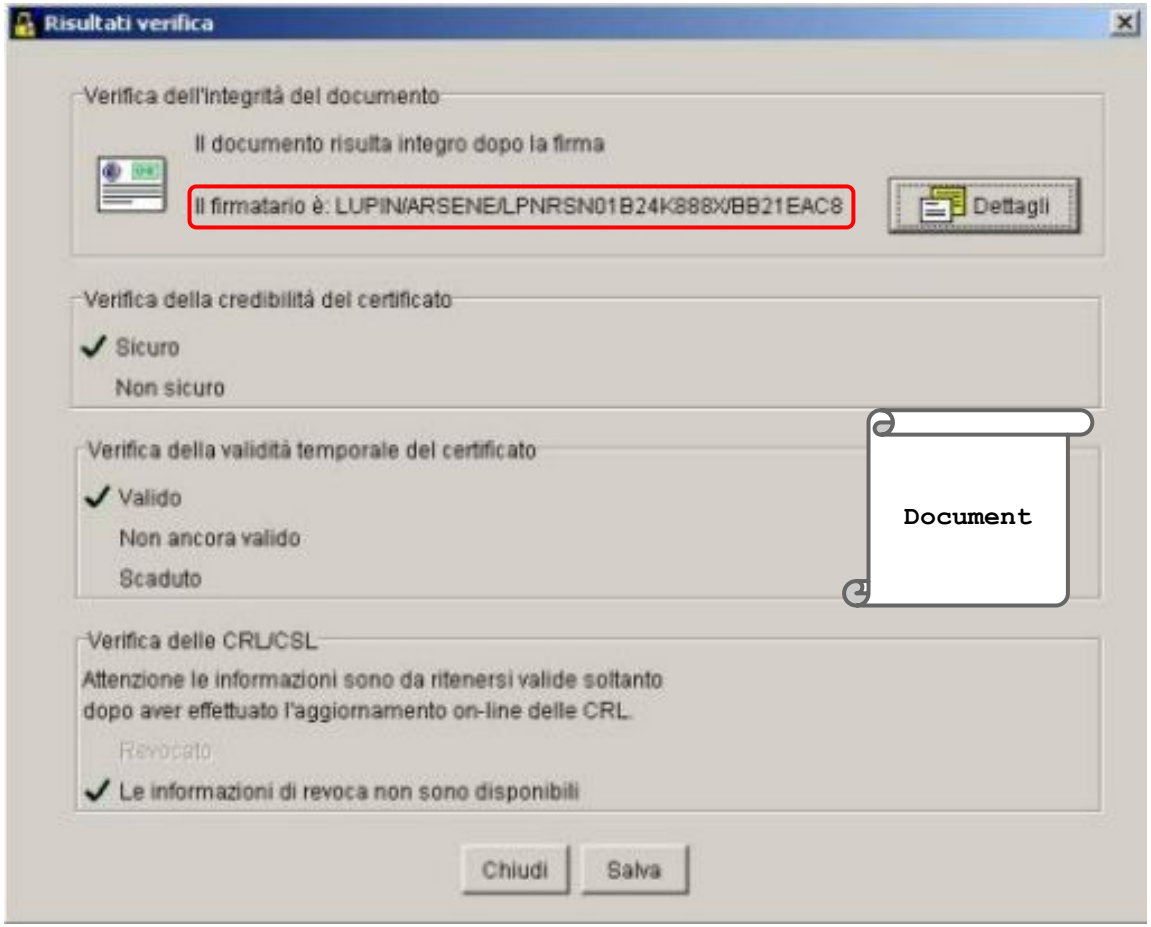
1. Generate a **fake *root* certificate** with the same name as a real one (e.g., PostECom itself)
2. Use this to generate a **fake *user* certificate** (in our example Arsène Lupin)
3. Use **Arsène Lupin's certificate** to sign theft and burglary confessions.
4. Include the fake root cert to the PKCS#7 envelope.



# Les jeux sont faits: Lupin's Certficate



# The Exploit: Arséne Lupin signature




Best comment by Postecom: this is *"by design"*
(yep: wrong design, but still design!)

# Conclusions

Perfect ciphers *vs.* real world: brute-forcing

- Broken-unbroken ciphers: need for transparency
- Key lengths matters
- Symmetric, asymmetric algorithms and hash functions
- PKI and CAs and their complexity

We saw several case studies of attacks against crypto applications

- They had everything to do with systems security without even touching the algorithms themselves

# Further reading: a practical attack based on MD5 collisions

- It is known that MD5 allows a **chosen prefix collision** under certain constraints
  - Here the attack is used to create **two valid CA certificates with the same signature**: http://www.win.tue.nl/~bdeweger/CollidingCertificates/
  - Extended to threaten CAs in 2008: http://www.win.tue.nl/hashclash/rogue-ca/
- An evolution of the technique was used in **Flame**, a nasty malware used against several middle-Eastern targets
  - http://trailofbits.files.wordpress.com/2012/06/flame-md5.pdf